

SPECTRAL GAPS FOR SETS AND MEASURES

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ABSTRACT. If X is a closed subset of the real line, denote by G_X the supremum of the size of the gap in the Fourier spectrum, taken over all non-trivial finite complex measures supported on X . In this paper we attempt to find G_X in terms of metric properties of X .

1. INTRODUCTION

Let μ be a non-zero finite complex measure on the real line. By $\hat{\mu}$ we denote its Fourier transform

$$\hat{\mu}(z) = \int \exp(-izt) d\mu(t).$$

Various properties of the Fourier transform of a measure have been studied by harmonic analysts for more than a century. One of the reasons for such a prolonged interest is the natural physical sense of the quantity $\hat{\mu}(t)$. For instance, in quantum mechanics, if μ is a spectral measure of a Hamiltonian then $|\hat{\mu}(t)|^2$ represents the so-called survival probability of the particle, i.e. the probability to find the particle in its initial state at the moment t . The problems considered in the present paper belong to the area of the Uncertainty Principle in Harmonic Analysis, whose name itself suggests relations and similarities with physics.

The Uncertainty Principle in Harmonic Analysis, as formulated in [12], says that a measure (function, distribution) and its Fourier transform cannot be simultaneously small. This broad statement gives rise to a multitude of exciting mathematical problems, each corresponding to a particular sense of "smallness."

One of such problems is the well-known Gap Problem. Here the smallness of μ and $\hat{\mu}$ is understood in the sense of porosity of their supports. The statement that one hopes to obtain is that if the support of $\hat{\mu}$ has a large gap then the support of μ cannot be too "rare." As usual, the ultimate challenge is to obtain quantitative estimates relating the two supports, something that we will attempt to do in this paper.

Beurling's Gap Theorem says that if the sequence of gaps in the support of μ is long, in the sense given by (5.2), then the support of $\hat{\mu}$ cannot have *any* gaps, unless μ is trivial, see [3]. The proof used some of the methods of an earlier gap theorem by Levinson [18]. In [7] de Branges proved that existence of a measure with a given spectral gap is equivalent to existence of a certain entire function of exponential type. We discuss a version of this result in section 8. For further results and references concerning the gap problem we refer the reader to [3, 2, 14, 18].

Our goal in this paper is to find an "if and only if" condition for a closed set X on the real line to support a non-trivial complex measure with a given size of

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the spectral gap. To this end we introduce a new metric characteristic of a closed set, C_X , see section 5 for the definition. Our main result is theorem 2 in section 6 that says that the supremum of the lengths of the spectral gaps, taken over all non-trivial measures supported on X , is equal to $2\pi C_X$.

The definition of C_X contains two conditions that, for the purposes of this paper, we call the density condition and the energy condition. As discussed in section 5, the density condition is similar to some of the definitions of densities used in this area. The physical flavor of the energy condition seems to suggest new connections for the gap problem that are yet to be fully understood.

As discussed in section 2, the gap problem can be equivalently reformulated as follows. Let μ be a finite complex measure on $\mathbb{R}$. Find the supremum, over all non-trivial $f \in L^1(|\mu|)$, of the size of the spectral gap of the measure $f\mu$. If one replaces L^1 with L^2 in this statement, one obtains (via simple duality) another famous problem, the problem of Wiener and Kolmogorov on completeness of families of exponentials in $L^2(\mu)$, see [16] or [10]. In [16] the problem is formulated in the language of Krein strings. Since for finite measures $L^2 \subset L^1$, our result gives an upper estimate for the Wiener-Kolmogorov problem. The L^∞ -version was considered by Koosis, see [14].

Via duality the gap problem admits a reformulation in terms of Bernstein's weighted approximation. From that point of view, $2\pi C_X$ is the minimal size of the interval such that continuous functions on X admit weighted approximation by trigonometric polynomials with frequencies from that interval, see section 3. The "approximative" relatives of the gap problem and connections with other classical areas, such as stationary Gaussian processes, are discussed in [11, 15, 14, 17].

Our methods are based on the approach developed by N. Makarov and the author in [19] and [20]. We utilize close connections between most problems from this area of harmonic analysis and the problem of injectivity of Toeplitz operators. In the case of the gap problem, this connection is expressed by theorem 1 below. The Toeplitz approach for similar problems was first suggested by Nikolski in [23], see also [24]. Our main proof utilizes several important ideas of the Beurling-Malliavin theory [4, 5, 20], including its famous multiplier theorem.

One of the advantages of the Toeplitz approach is that it reveals hidden connections between various problems of analysis and mathematical physics, see [19]. The relations between the gap problem and the Beurling-Malliavin theory on completeness of exponentials in L^2 on an interval have been known to experts, at a rather intuitive level, for several decades. Now we can see this connection formulated in precise mathematical terms. Namely, the Beurling-Malliavin problem is equivalent to the problem of triviality of the kernel of a Toeplitz operator with the symbol

$$\phi = \exp(-iax)\theta,$$

for a suitable meromorphic inner function θ , whereas the gap problem is equivalent to the triviality of the kernel of the Toeplitz operator with the symbol

$$\bar{\phi} = \exp(iax)\bar{\theta},$$

see [19] section 4.6 and theorem 1 below.

The paper is organized as follows:

- In section 2 we discuss an alternative formulation of the gap problem and show that the maximal size of the gap for a fixed measure taken over all possible densities is a property of the support of the measure.
- In section 3 we look at the gap problem from the point of view of Bernstein's weighted approximation of continuous functions by trigonometric polynomials.
- In section 4 we restate the gap problem in terms of kernels of Toeplitz operators and introduce the approach that will be used in the main proof.
- Section 5 contains the main definition and its discussion. For a closed real set X we define a metric characteristic C_X that determines the maximal size of the gap over all non-zero complex measures supported on X .
- Section 6 contains the main result and its proof.
- In section 7 we prove several technical lemmas and corollaries used in sections 5 and 6.
- Section 8 can be viewed as an appendix. It contains a Toeplitz version of the statement and proof of theorem 66 from [7].

2. SPECTRAL GAP AS A PROPERTY OF THE SUPPORT

Let M be a set of all finite Borel complex measures on the real line. If X is a closed subset of the real line denote

$$G_X = \sup\{a \mid \exists \mu \in M, \mu \neq 0, \text{supp } \mu \subset X, \text{ such that } \hat{\mu} = 0 \text{ on } [0, a]\}. \quad (2.1)$$

Now let $\mu \in M$. Denote

$$G_\mu = \sup\{a \mid \exists f \in L^1(|\mu|) \text{ such that } \widehat{f\mu} = 0 \text{ on } [0, a]\}. \quad (2.2)$$

Proposition 1.

$$G_\mu = G_{\text{supp } \mu}. \quad (2.3)$$

Proof. Obviously, $G_{\text{supp } \mu} \geq G_\mu$. To prove the opposite inequality, notice that by lemma 8 there exists a finite discrete measure

$$\nu = \sum \alpha_n \delta_{x_n}, \{x_n\} \subset \text{supp } \mu,$$

such that $\hat{\nu}$ has a gap of the size greater than $G_{\text{supp } \mu} - \varepsilon$. Around each x_n choose a small neighborhood $V_n = (a_n, b_n)$ so that for any sequence of points

$$Y = \{y_n\}, y_n \in V_n$$

there exists a non-trivial measure $\eta_Y = \sum \beta_n \delta_{y_n}$ such that $\hat{\eta}$ has a gap of the size greater than $G_{\text{supp } \eta} - \varepsilon$. The existence of such a collection of neighborhoods follows from the results of [2] (for some sequences), from [6] as well as from theorem 2 below.

Now one can choose a family of finite measures $\eta_\tau, \tau \in [0, 1]$ with the following properties:

- for each τ ,

$$\eta_\tau = \sum \beta_n^\tau \delta_{y_n^\tau}$$

such that $y_n^\tau \in V_n$ and $\hat{\eta}_\tau$ has a gap of the size greater than $G_{\text{supp } \eta} - \varepsilon$ centered at 0;

- the measure

$$\gamma = \int_0^1 \eta_\tau d\tau$$

is non-trivial and absolutely continuous with respect to μ .

It remains to notice that then $\hat{\gamma}$ has a gap of the size greater than $G_{\text{supp } \eta} - \varepsilon$. $\square$

3. APPROXIMATION OF CONTINUOUS FUNCTIONS BY TRIGONOMETRIC POLYNOMIALS: BERNSTEIN'S VERSION OF THE GAP PROBLEM

Definition 1. Let $X \subset \mathbb{R}$ be a closed set. A weight is any lower semicontinuous function $W : \mathbb{R} \rightarrow [1, \infty)$ that tends to ∞ as $x \rightarrow \pm\infty$. For any given weight W we define $C_0(W, X)$ to be the space of all continuous functions on X satisfying

$$\lim_{x \in X, x \rightarrow \pm\infty} \frac{f(x)}{W(x)} = 0.$$

(If X is bounded from below or from above, the corresponding limit is dropped from the definition. In particular, for bounded X , $C_0(W, X)$ is just $C(X)$.)

We define the norm in $C_0(W, X)$ as

$$\|f\| = \|fW^{-1}\|_\infty.$$

As usual, we say that a system of functions is complete in a space if finite linear combinations of functions from that system are dense in the space.

If $a > 0$ we denote by $\mathcal{E}_a$ the set of complex exponentials with frequencies between 0 and a :

$$\mathcal{E}_a = \{ \exp(i\lambda t) \mid \lambda \in [0, a] \}.$$

Definition 2. If $X \subset \mathbb{R}$ is a closed set, define the approximative capacity of X , A_X , as

$$A_X = \inf \{ a \mid \mathcal{E}_a \text{ is complete in } C_0(W, X) \text{ for any weight } W \}$$

or ∞ if the set is empty.

The same quantity can be defined in a different way. Denote by $C_0(X)$ the space of all continuous functions on X tending to 0 at infinity, with the usual sup-norm. If one now wants to discuss approximation by trigonometric polynomials in $C_0(X)$, he encounters a small problem: exponential functions are no longer inside the space. The solution is to consider "linear combinations" of exponentials, e.g. the Payley-Wiener space

$$PW_a = \{ \hat{f} \mid f \in L^2([0, a]) \},$$

and define

$$A_X = \inf\{a \mid PW_a \text{ is dense in } C_0(X)\}$$

or ∞ if the set is empty.

It is not difficult to show that the above definitions of A_X are equivalent.

The following statement is a product of the standard duality argument.

Proposition 2.

$$A_X = G_X.$$

Together with theorem 2, this statement gives a formula for A_X .

4. MODEL SPACES AND CLARK MEASURES. THE TOEPLITZ VERSION OF THE GAP PROBLEM

By H^2 we denote the Hardy space in the upper half-plane $\mathbb{C}_+$. We say that an inner function $\theta(z)$ in $\mathbb{C}_+$ is meromorphic if it allows a meromorphic extension to the whole complex plane. The meromorphic extension to the lower half-plane $\mathbb{C}_-$ is given by

$$\theta(z) = \frac{1}{\theta^\#(z)}.$$

Each inner function $\theta(z)$ determines a model subspace

$$K_\theta = H^2 \ominus \theta H^2$$

of the Hardy space $H^2(\mathbb{C}_+)$. These subspaces play an important role in complex and harmonic analysis, as well as in operator theory, see [24].

Each inner function $\theta(z)$ determines a positive harmonic function

$$\Re \frac{1 + \theta(z)}{1 - \theta(z)}$$

and, by the Herglotz representation, a positive measure σ such that

$$\Re \frac{1 + \theta(z)}{1 - \theta(z)} = py + \frac{1}{\pi} \int \frac{y d\sigma(t)}{(x - t)^2 + y^2}, \quad z = x + iy, \quad (4.1)$$

for some $p \geq 0$. The number p can be viewed as a point mass at infinity. The measure σ is singular, supported on the set where non-tangential limits of θ are equal to 1 and satisfies

$$\int \frac{d\sigma(t)}{1 + t^2} < \infty. \quad (4.2)$$

The measure $\sigma + p\delta_\infty$ on $\hat{\mathbb{R}}$ is called the Clark measure for $\theta(z)$. (Following standard notations, we will often denote the Clark measure defined in (4.1) by σ_1 .)

Conversely, for every positive singular measure σ satisfying (4.2) and a number $p \geq 0$, there exists an inner function $\theta(z)$ determined by the formula (4.1).

Every function $f \in K_\theta$ can be represented by the formula

$$f(z) = \frac{p}{2\pi i} (1 - \theta(z)) \int f(t) \overline{(1 - \theta(t))} dt + \frac{1 - \theta(z)}{2\pi i} \int \frac{f(t)}{t - z} d\sigma(t). \quad (4.3)$$

If the Clark measure does not have a point mass at infinity, the formula is simplified to

$$f(z) = \frac{1}{2\pi i} (1 - \theta(z)) K f \sigma$$

where $K f \sigma$ stands for the Cauchy integral

$$K f \sigma(z) = \int \frac{f(t)}{t - z} d\sigma(t).$$

This gives an isometry of $L^2(\sigma)$ onto K_θ . In the case of meromorphic $\theta(z)$, every function $f \in K_\theta$ also has a meromorphic extension in $\mathbb{C}$, and it is given by the formula (4.3). The corresponding Clark measure is discrete with atoms at the points of $\{\theta = 1\}$ given by

$$\sigma(\{x\}) = \frac{2\pi}{|\theta'(x)|}.$$

For more details on Clark measures the reader may consult [26] or the references therein.

Each meromorphic inner function $\theta(z)$ can be written as $\theta(t) = e^{i\phi(t)}$ on $\mathbb{R}$, where $\phi(t)$ is a real analytic and strictly increasing function. The function $\phi(t) = \arg \theta(t)$ is the continuous argument of $\theta(z)$.

Recall that the Toeplitz operator T_U with a symbol $U \in L^\infty(\mathbb{R})$ is the map

$$T_U : H^2 \rightarrow H^2, \quad F \mapsto P_+(UF),$$

where P_+ is the orthogonal projection in $L^2(\mathbb{R})$ onto the Hardy space $H^2 = H^2(\mathbb{C}_+)$.

We will use the following notation for kernels of Toeplitz operators (or *Toeplitz kernels* in H^2):

$$N[U] = \ker T_U.$$

For example, $N[\bar{\theta}] = K_\theta$ if θ is an inner function. Along with H^2 -kernels, one may consider Toeplitz kernels $N^p[U]$ in other Hardy classes H^p , the kernel $N^{1,\infty}[U]$ in the "weak" space $H^{1,\infty} = H^p \cap L^{1,\infty}$, $0 < p < 1$, or the kernel in the Smirnov class $N^+(\mathbb{C}_+)$:

$$N^+[U] = \{f \in N^+ \cap L^1_{loc}(\mathbb{R}) : \bar{U}\bar{f} \in N^+\}.$$

For more on such kernels see [19, 20].

For any inner function θ in the upper half-plane we denote by spec_θ the set $\{\theta = 1\}$, the set of points on the line where the non-tangential limit of θ is equal to 1, plus the infinite point if the corresponding Clark measure has a point mass at infinity, i.e. if p in (4.1) is positive. If $\text{spec}_\theta \subset \mathbb{R}$ like in the next definition, then p in (4.1) is 0. Throughout the paper, S stands for the exponential inner function $S(z) = \exp(iz)$.

Definition 3. If $X \subset \mathbb{R}$ is a closed set, denote

$$T_X = \sup \{ a \mid N[\bar{\theta}S^a] \neq 0 \text{ for some meromorphic inner } \theta, \text{spec}_\theta \subset X \}.$$

The following theorem shows the connection between the gap problem and the problem of triviality of Toeplitz kernels. This connection will be used throughout the paper.

Theorem 1.

$$T_X = G_X.$$

We call a sequence of real points discrete if it has no finite accumulation points. Note that spec_θ is discrete if and only if θ is meromorphic.

Proof. Let $T_X = d$. Then for any $\varepsilon > 0$ there exists a discrete sequence $\Lambda \subset X$ such that the kernel $N[\bar{\theta}S^a]$ is non-trivial for some/any meromorphic inner θ , $\text{spec}_\theta = \Lambda$ and $a = d - \varepsilon$ (if the kernel is non-trivial for some θ_1 with $a = d - \varepsilon/2$ then it is non-trivial for any θ_2 , $\text{spec}_{\theta_1} = \text{spec}_{\theta_2}$, with $a = d - \varepsilon$, see [19]). Let $f \in N[\bar{\theta}S^a]$. Then $h = S^a f \in K_\theta$ and by the Clark formula

$$h = \frac{1}{2\pi i}(1 - \theta)Kh\sigma_1,$$

where σ_1 is the Clark measure corresponding to θ . Notice that the function $1 - \theta$ decays at most as y^{-1} along the positive y -axis. Hence the Cauchy integral $Kh\sigma_1$ decays as $\exp(-ay)$ as $y \rightarrow \infty$. Hence the measure $\nu = h\sigma_1$ satisfies $\text{supp } \nu = \text{supp } \sigma_1 = \text{spec}_\theta \subset X$.

The function θ can be chosen so that σ_1 is finite: one can start by choosing any finite positive σ_1 supported by Λ and then simply take θ corresponding to that measure. Then ν is finite as well. It remains to observe that ν has a spectral gap of the size at least $d - \varepsilon$ and $G_X \geq d$ (see, for instance, lemma 7).

In the opposite direction, if $G = d$ then, by lemma 4, for any $\varepsilon > 0$ there exists a finite measure ν with a spectral gap at least $d - \varepsilon$ concentrated on a discrete subset of X . Let θ be the inner function corresponding to $|\nu|$. Then the function $h = (1 - \theta)K\nu$ belongs to K_θ and can be represented as $h = S^{a-\varepsilon}f$ for some $f \in H^2$. Hence $f \in N[\bar{\theta}S^{a-\varepsilon}]$ and $T_X \geq d$. $\square$

5. THE MAIN DEFINITION

Let $\Lambda = \{\lambda_1, \dots, \lambda_n\}$ be a finite set of points on $\mathbb{R}$. Consider the quantity

$$E(\Lambda) = \sum_{\lambda_k, \lambda_j \in \Lambda} \log |\lambda_k - \lambda_j|. \quad (5.1)$$

Physical interpretation:

According to the 2-dimensional Coulomb's law, $E(\Lambda)$ is the energy of a system of "flat" electrons placed at the points of Λ . The 2D Coulomb-gas formalism corresponds to the planar potential theory with logarithmic potential and assumes the potential energy at infinity to be equal to $-\infty$, see for instance [9, 22, 27].

Physically, the 2D Coulomb's law can be derived from the standard 3D law via a method of "reduction." According to this method, one replaces each electron in the plane with a uniformly charged string orthogonal to the plane. After that one applies the 3D law and a renormalization procedure.

Key example:

Let $I \subset \mathbb{R}$ be an interval, $\Lambda = I \cap \mathbb{Z} = \{n+1, \dots, n+k\}$. Then

$$E(\Lambda) = k^2 \log |I| + O(|I|^2)$$

as follows from Stirling's formula. Here $|I|$ stands for the length of I and the notation $O(|I|^2)$ corresponds to the direction $|I| \rightarrow \infty$.

Remark 1. *The uniform distribution of points on the interval does not maximize the energy $E(\Lambda)$, but comes within $O(|I|^2)$ from the maximum, which is negligible for our purposes, see the main definition and its discussion below. It is interesting to observe that the maximal energy for k points is achieved when the points are placed at the endpoints of I and the zeros of the Jacobi polynomial of degree $k - 2$, see for instance [13].*

We call a sequence of disjoint intervals $\{I_n\}$ on the real line long (in the sense of Beurling and Malliavin) if

$$\sum_n \frac{|I_n|^2}{1 + \text{dist}^2(0, I_n)} = \infty. \quad (5.2)$$

If the sum is finite we call $\{I_n\}$ short.

Let

$$\dots < a_{-2} < a_{-1} < a_0 = 0 < a_1 < a_2 < \dots$$

be a two-sided sequence of real points. We say that the intervals $I_n = (a_n, a_{n+1}]$ form a short partition of $\mathbb{R}$ if $|I_n| \rightarrow \infty$ as $n \rightarrow \pm\infty$ and the sequence $\{I_n\}$ is short.

Main definition:

Let $\Lambda = \{\lambda_n\}$ be a sequence of real points. We write $C_\Lambda \geq a$ if there exists a short partition $\{I_n\}$ such that

$$\Delta_n \geq a|I_n| \quad \text{for all } n \quad (\text{density condition}) \quad (5.3)$$

and

$$\sum_n [\Delta_n^2 \log |I_n| - E_n] / (1 + \text{dist}^2(0, I_n)) < \infty \quad (\text{energy condition}) \quad (5.4)$$

where

$$\Delta_n = \#(\Lambda \cap I_n) \quad \text{and} \quad E_n = E(\Lambda \cap I_n) = \sum_{\lambda_k, \lambda_l \in I_n, \lambda_k \neq \lambda_l} \log |\lambda_k - \lambda_l|.$$

If X is a closed subset of $\mathbb{R}$ we put

$$C_X = \sup\{a \mid \text{there exists a sequence } \Lambda \subset X \text{ such that } C_\Lambda \geq a\} \quad (5.5)$$

Remark 2. *Notice that the series in the energy condition is positive. Indeed, every term in the sum defining E_n is at most $\log |I_n|$ and there are precisely Δ_n^2 terms.*

The example before the definition shows that the numerator in the energy condition is (up to lower order terms) the difference between the energy of the optimal configuration, when the points are spread uniformly on I_n , and the energy of $\Lambda \cap I_n$. Thus the energy condition is a requirement that the placement of the points of Λ is close to uniform, in the sense that the difference between the energies must be summable with respect to the Poisson weight.

Remark 3. The simple inequality $\log |I_n| - \log |\lambda_k - \lambda_j| > 0$, that holds for any $\lambda_k, \lambda_j \in I_n$, also implies that if a sequence Λ satisfies the energy condition (5.4) then any subsequence of Λ also satisfies (5.4) on $\{I_n\}$. A deletion of points from Λ will eliminate some of such positive terms from the numerator in (5.4) which can only make the sum smaller.

Remark 4. We say that a partition $\{I_n\}$ is monotone if $|I_n| \leq |I_{n+1}|$ for $n \geq 0$ and $|I_{n+1}| \leq |I_n|$ for $n < 0$. Corollary 3 shows that in the above definition the words "short partition" can be replaced with "short monotone partition." Since monotone partitions are easier to work with, this modified definition will be used in the proof of theorem 2.

Remark 5. The requirement that the partition $I_n = (a_n, a_{n+1}]$ satisfied $|I_n| \rightarrow \infty$ is not essential and can be omitted if one slightly changes the definitions of Δ_n and E_n in (5.4). One could, for instance, use

$$\Delta_n = \#(\Lambda \cap (a_n - 1, a_{n+1} + 1]) \quad \text{and} \quad E_n = \sum_{\lambda_k, \lambda_l \in (a_n - 1, a_{n+1} + 1], \lambda_k \neq \lambda_l} \log |\lambda_k - \lambda_l|.$$

Remark 6. The density condition says simply that the lower (interior) density of the sequence in the sense of Beurling and Malliavin, is at least a . Such a density can be defined in several different ways:

- If Λ is a real sequence define $d_1(\Lambda)$ to be the supremum of all a such that there exists a short monotone partition I_n satisfying (5.3).
- Denote by $d_2(\Lambda)$ the supremum of all a such that there exists a short (not necessarily monotone) partition I_n satisfying (5.3).
- Define $d_3(\Lambda)$ to be the supremum of all a such that there exists a subsequence of Λ whose counting function $n(x)$ satisfies

$$\int \frac{|n - ax|}{1 + x^2} dx < \infty.$$

This notion of lower density is used in [7].

- Finally, define $d_4(\Lambda)$ to be the infimum of all a such that there exists a long sequence of disjoint intervals I_n satisfying

$$\#\Lambda \cap I_n < a|I_n|.$$

This definition is used in [21].

One can easily show that all these definitions are equivalent, i.e.

$$d_1(\Lambda) = d_2(\Lambda) = d_3(\Lambda) = d_4(\Lambda).$$

Such a density (with a definition similar to the last one) was introduced in [5], where it was called interior density. A closely related notion of exterior density appears in the Beurling-Malliavin Theorem on completeness of exponential functions in L^2 on an interval, see [5] or [21].

Example 1. As discussed above, if the points of the sequence are spread uniformly over the interval then $E_n = \sum_{\lambda_i, \lambda_j \in I_n} \log |\lambda_i - \lambda_j|$ is roughly (up to $O(|I_n|^2)$ which is small for short sequences of I_n) equal to $\Delta_n^2 \log |I_n|$ as follows from Stirling's formula. This happens for instance when the sequence Λ is separated, i.e. satisfies $|\lambda_n - \lambda_{n+1}| > \delta > 0$ for all n . Thus for separated sequences Λ the energy condition disappears and

$$G_\Lambda = d_i(\Lambda)$$

where $d_i, i = 1, 2, 3, 4$ is any of the equivalent densities defined in the previous remark. This is one of the results of [21].

Example 2. Let Λ be a real sequence such that the density condition (5.3) holds for some $a > 0$ and some partition $\{I_n\}$ that satisfies a stronger shortness condition:

$$\sum_n \frac{|I_n|^2 \log |I_n|}{1 + \text{dist}^2(0, I_n)} < \infty.$$

Then we will automatically have that

$$\sum_n \frac{\Delta_n^2 \log |I_n| - \sum_{\lambda_i, \lambda_j \in I_n} \log_+ |\lambda_i - \lambda_j|}{1 + \text{dist}^2(0, I_n)} < \infty.$$

Hence condition (5.4) will be significantly simplified and one will only need to check that

$$\sum_{\lambda_i, \lambda_j \in \Lambda, \lambda_i \neq \lambda_j} \frac{\log_- |\lambda_i - \lambda_j|}{1 + \lambda_j^2} < \infty$$

to conclude that $C_\Lambda \geq a$.

6. THE MAIN THEOREM

Theorem 2.

$$G_X = 2\pi C_X.$$

Before starting the proof, let us discuss some of the notations. If f is a function on $\mathbb{R}$ and $I \subset \mathbb{R}$ we denote by $f|_I$ the function that is equal to f on I and to 0 on $\mathbb{R} \setminus I$.

In our estimates we write $a(n) \lesssim b(n)$ if $a(n) < Cb(n)$ for some positive constant C , not depending on n , and large enough $|n|$. Similarly, we write $a(n) \asymp b(n)$ if $ca(n) < b(n) < Ca(n)$ for some $C \geq c > 0$. Some formulas will have other parameters in place of n or no parameters at all.

By Π we denote the Poisson measure $dx/(1+x^2)$ on the real line. In particular, $L_\Pi^p = L^p(\mathbb{R}, dx/(1+x^2))$.

We will denote by $\mathcal{D}(\mathbb{R})$ the standard Dirichlet space on $\mathbb{R}$ (in $\mathbb{C}_+$). Recall that the Hilbert space $\mathcal{D}(\mathbb{R})$ consists of functions $h \in L_\Pi^1$ such that the harmonic extension $u = u(z)$ of h to $\mathbb{C}_+$ has a finite gradient norm,

$$\|h\|_{\mathcal{D}}^2 \equiv \|u\|_{\nabla}^2 \stackrel{\text{def}}{=} \int_{\mathbb{C}_+} |\nabla u|^2 dA < \infty,$$

where dA is the area measure. If $h \in \mathcal{D}(\mathbb{R})$ is a smooth function, then we also have

$$\|h\|_{\mathcal{D}}^2 = \int_{\mathbb{R}} \bar{h} \tilde{h}' dx,$$

where $\tilde{h}$ denotes a harmonic conjugate function.

Proof:

I) First suppose that $C_X > \frac{1}{2\pi}$. We will show that $G_X \geq 1$.

Choose $\varepsilon > 0$. If $C_X > \frac{1}{2\pi}$, there exists a sequence $\Lambda = \{\lambda_n\} \subset X$, $C_\Lambda > \frac{1}{2\pi}$. Let

$$I_n = (a_n, a_{n+1})$$

be the corresponding short monotone partition, see remark 4. WLOG

$$\frac{1}{2\pi}|I_n| < \#(\Lambda \cap I_n) \leq \frac{1}{2\pi}|I_n| + 1$$

(otherwise just delete some of the points from Λ , see remark 3). We will assume that $|I_n| \gg 1/\varepsilon \gg 1$ for all n .

By lemma 1 and corollary 2 we can assume that the lengths of the intervals $(\lambda_n, \lambda_{n+1})$ are bounded from above. It will be convenient for us to assume that the endpoints of I_n belong to Λ , i.e. that $I_n = (\lambda_{k_n}, \lambda_{k_{n+1}}]$ for some $\lambda_{k_n}, \lambda_{k_{n+1}} \in \Lambda$. We will also include the endpoints of the intervals into the energy condition by defining E_n as

$$E_n = \sum_{\lambda_{k_n} \leq \lambda_k, \lambda_l \leq \lambda_{k_{n+1}}, \lambda_k \neq \lambda_l} \log |\lambda_k - \lambda_l| \quad (6.1)$$

and assuming that (5.4) is satisfied with these E_n . Such an assumption can be made because if the sum in (5.4) becomes infinite with E_n defined by (6.1) one can, for instance, delete the first point, $\lambda_{k_{n+1}}$, from Λ on all I_n for large n . After the addition of λ_{k_n} and deletion of $\lambda_{k_{n+1}}$ in the sum defining E_n , each term in (5.4) will become smaller and the sum will remain finite. At the same time, since

$$|I_n| \asymp \#\{\Lambda \cap I_n\} \rightarrow \infty,$$

the subsequence will still have more than $|I_n|$ points on each I_n and will satisfy the density condition.

Our goal is to show that $G_\Lambda \geq 1$ by producing a measure on Λ with spectral gap of the size arbitrarily close to 1. Due to connections discussed in section 4, existence of such a measure will follow from non-triviality of a certain Toeplitz kernel.

Since the lengths of $(\lambda_n, \lambda_{n+1})$ are bounded from above, we can apply lemma 5. Denote by θ the corresponding meromorphic inner function with $\text{spec}_\theta = \Lambda$.

Let $u = \arg(\theta \bar{S}) = \arg \theta - x$. First, we choose a larger partition $J_n = (b_n, b_{n+1})$ and a small "correction" function v so that $u - v$ becomes an atom on each J_n :

Claim 1. *There exists a subsequence $\{b_n\}$ of the sequence $\{a_n\}$ and smooth functions v_1, v_2 such that:*

- 1) $|v'_1| < \varepsilon/2$ and $u - v_1 = 0$ at all a_n ;
- 2) $J_n = (b_n, b_{n+1})$ is a short monotone partition;
- 3) $|v'_2| < \varepsilon/2$ and $u - v = u - (v_1 + v_2) = 0$ at all b_n ;
- 4) $\int_{J_n} (u - v) dx = 0$ for all n ;

$$5) \tilde{u} - \tilde{v} \in L_{\Pi}^1.$$

Proof of claim. First, choose a smooth function v_1 satisfying 1. Such a function exists because

$$|2\pi\Delta_n - |I_n|| \leq 2\pi < \frac{\varepsilon}{2}|I_n|.$$

Notice that because the sequence I_n is short and

$$(u - v_1)' > -1 - \frac{\varepsilon}{2},$$

1 implies

$$u - v_1 \in L_{\Pi}^1. \quad (6.2)$$

Choose $b_0 = a_0 = 0$. Choose $b_1 = a_{n_1} > b_0$ to be the smallest element of $\{a_k\}$ satisfying

$$\left| \int_{b_0}^{a_{n_1}} (u - v_1) dx \right| < \frac{\varepsilon}{8} (a_{n_1} - b_0)^2.$$

Notice that because of (6.2) such an a_{n_1} will always exist. After that proceed choosing $b_2, b_3, \dots$ in the following way: If b_i is chosen, choose $b_{i+1} = a_{n_{i+1}}$ to be the smallest element of $\{a_k\}$ satisfying $a_{n_{i+1}} > b_i$,

$$\left| \int_{b_i}^{a_{n_{i+1}}} (u - v_1) dx \right| < \frac{\varepsilon}{8} (a_{n_{i+1}} - b_i)^2 \quad (6.3)$$

and

$$a_{n_{i+1}} - b_i \geq b_i - b_{i-1}.$$

Choose $b_k, k < 0$ in the same way.

We claim that the resulting sequence $J_k = (b_{k-1}, b_k)$ forms a short monotone partition.

Let k be positive. By our construction, I_{n_k} is the last (rightmost) among the intervals I_n contained in J_k . Notice that because of monotonicity I_{n_k} is the largest interval among the intervals I_n contained in J_k . We will show that for each k

$$|J_k| < \left(\left\lceil \frac{10}{\varepsilon} \right\rceil + 1 \right) |I_{n_k}| \quad (6.4)$$

where $\lceil \cdot \rceil$ stands for the entire part of a real number.

This can be proved by induction. The basic step: By our construction $b_1 = a_{n_1}$ and

$$\left| \int_{b_0}^{a_{n_1-1}} (u - v_1) dx \right| \geq \frac{\varepsilon}{8} (a_{n_1-1} - b_0)^2.$$

Since $(u - v_1)' > -1 - \varepsilon$ and $u - v_1 = 0$ at all a_n , $|u - v_1| \leq (1 + \varepsilon)|I_{n_1-1}|$ on (b_0, a_{n_1-1}) . Hence

$$(1 + \varepsilon)|I_{n_1-1}|(a_{n_1-1} - b_0) \geq \left| \int_{b_0}^{a_{n_1-1}} (u - v_1) dx \right| \geq \frac{\varepsilon}{8} (a_{n_1-1} - b_0)^2$$

and

$$(a_{n_1-1} - b_0) \leq 8 \frac{1 + \varepsilon}{\varepsilon} |I_{n_k}|.$$

It follows that

$$|J_1| = (a_{n_1-1} - b_0) + |I_{n_1}| \leq 9\varepsilon^{-1}|I_{n_1-1}| + |I_{n_1-1}| \leq \frac{10}{\varepsilon}|I_{n_1-1}| \quad (6.5)$$

(if ε is small enough). For the inductive step, assume that (6.4) holds for $k = l - 1$. For $J_l = (b_{l-1}, b_l)$, $b_l = a_{n_l}$ there are two possibilities:

$$\left| \int_{b_{l-1}}^{a_{n_l-1}} (u - v_1) dx \right| \geq \frac{\varepsilon}{8} (a_{n_l-1} - b_{l-1})^2$$

or

$$a_{n_l-1} - b_{l-1} < b_{l-1} - b_{l-2}.$$

In the first case we prove (6.5) in the same way as in the basic step. In the second case we notice that by monotonicity of I_n the number of intervals I_n inside (b_{l-1}, a_{n_l-1}) is at most $(a_{n_l-1} - b_{l-1})/|I_{n_l-1}|$ which is strictly less than $|J_{l-1}|/|I_{n_l-1}| \leq [10/\varepsilon] + 1$. Hence the number of intervals in (b_{l-1}, a_{n_l-1}) is at most $[10/\varepsilon]$. Therefore the number of intervals in $J_l = (b_{l-1}, b_l)$ is at most $[10/\varepsilon] + 1$. Now, since I_{n_l} is the largest interval in J_l we again get (6.4), which implies shortness of J_n . The monotonicity follows from our construction.

Now define the function v_2 on each J_k in the following way. First consider the tent function T_k defined on $\mathbb{R}$ as

$$T_k(x) = \frac{\varepsilon}{4} \text{dist}(x, \mathbb{R} \setminus J_k).$$

Notice that because of (6.3), for each k there exists a constant C_k , $|C_k| \leq 1$ such that

$$\int_{J_k} [(u - v_1) - C_k T_k] dx = 0.$$

Now define v_2 as a smoothed-out sum $\sum C_k T_k$ that satisfies $|v_2'| < \varepsilon/2$ and still has the properties that $v_2(b_k) = 0$ and

$$\int_{J_k} [(u - v_1) - v_2] dx = 0$$

for each k . Finally, let $v = v_1 + v_2$. The last condition of the claim will be satisfied because the restrictions $(u - v)|_{J_k}$ form a collection of atoms with a finite sum of L^1_Π -norms:

$$\|(u - v)|_{J_k}\|_{L^1_\Pi} \lesssim \frac{|J_k|^2}{1 + \text{dist}^2(0, J_k)}$$

(for more on atomic decompositions see [8]). $\triangle$

The function v from the last claim is a smooth function satisfying $|v'| \leq \varepsilon$. Therefore it can be represented as $v = v_+ - v_-$ where $v_\pm$ are smooth growing functions, $0 \leq v'_\pm \leq \varepsilon$. Hence one can choose two meromorphic inner functions $I_\pm$ satisfying

$$\{\arg I_\pm = k\pi\} = \{\arg v_\pm = k\pi\}$$

and

$$|I'_\pm| \lesssim \varepsilon$$

(the existence of such $I_{\pm}$ follows, for instance, from lemma 5 or from the lemma in section 6.1 of [20]).

Note that then, automatically, $|\arg(\bar{I}_+ I_-) - v| < 2\pi$. The harmonic conjugate of $\arg(\theta \bar{S} I_+ \bar{I}_-)$ still belongs to L^1_{Π} .

WLOG $\arg(\theta \bar{S} I_+ \bar{I}_-) = 0$ at 0.

Claim 2. *The function $\arg(\theta(x) \bar{S} I_+(x) \bar{I}_-(x))/x$ belongs to the Dirichlet class $\mathcal{D}(\mathbb{R})$.*

Proof of claim. We will actually prove that w/x , $w = \arg \theta - x - v \in \mathcal{D}$ instead (again, WLOG $w(0) = 0$ with large multiplicity). The difference between $-v$ and $\arg(I_+ \bar{I}_-)$ is a bounded function with bounded derivative that obviously belongs to $\mathcal{D}$.

Let $q(z)$ be the harmonic extension of w/x to the upper half plane. We need to show that the gradient norm of $q + i\tilde{q}$ in $\mathbb{C}_+$ is finite, i.e. that

$$\|q + i\tilde{q}\|_{\nabla}^2 = \lim_{r \rightarrow \infty} \int_{\partial D(r)} q d\tilde{q} < \infty,$$

where $D(r)$ is the semidisc $\{|z| < r\} \cap \mathbb{C}_+$.

We first prove that the integrals over $\partial D(r) \cap \mathbb{R}$ are uniformly bounded from above, i.e. that

$$\int_{\mathbb{R}} q d\tilde{q} < \infty.$$

First, notice that the harmonic conjugate of $\frac{w}{x}$ is $\frac{\tilde{w}}{x}$ and $(\frac{w}{x})' = \frac{w'}{x} - \frac{w}{x^2}$, where $\frac{w}{x^2}$ is a bounded function since w' is bounded and has a zero at zero. Hence

$$\int_{\mathbb{R}} q d\tilde{q} \asymp \int_{\mathbb{R}} w' \tilde{w} \frac{dx}{x^2}$$

and we can estimate the last integral instead.

If I is an interval then $2I$ denotes the interval with the same center as I satisfying $|2I| = 2|I|$.

Put $w_n = w|_{J_n}$. Then

$$\int_{\mathbb{R}} w' \tilde{w} \frac{dx}{x^2} = \sum_n \sum_k \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2}.$$

To estimate the last integral, first let us consider the case when the intervals J_n and J_k are far from each other:

$$\max(|J_n|, |J_k|) \leq \text{dist}(J_n, J_k).$$

In this case

$$\begin{aligned} \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} &\lesssim \int_{J_n} |w'| \frac{|J_k|^3}{\text{dist}^2(J_k, x)} \frac{dx}{x^2} \lesssim \\ &\frac{|J_k|^3}{1 + \text{dist}^2(J_n, 0)} \int_{J_n} \frac{dx}{\text{dist}^2(J_k, x)}. \end{aligned} \quad (6.6)$$

Here we used the property that each w_k is an atom supported on J_k whose L^1 -norm is $\lesssim |J_k|^2$ and employed the standard estimates from the theory of atomic decompositions, see [8]. In the last inequality we used the property

$$\int_{J_n} |w'(x)| dx \lesssim |J_n|. \quad (6.7)$$

Now let us consider the "mid-range" case when

$$\min(|J_n|, |J_k|) \leq \text{dist}(J_n, J_k) < \max(|J_n|, |J_k|).$$

Assume that $0 < k < n$. Then by monotonicity $|J_k| \leq |J_n|$. In this case

$$\begin{aligned} \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} &\lesssim \frac{|J_k|^2}{1 + \text{dist}^2(J_k, 0)} \int_{J_n} |w'| \frac{|J_k|}{\text{dist}^2(J_k, x)} dx \leq \\ &\frac{|J_k|^2}{1 + \text{dist}^2(J_k, 0)} \frac{|J_k|}{\text{dist}^2(J_k, J_n)} \int_{J_n} |w'| \lesssim \frac{|J_k| |J_n|}{1 + \text{dist}^2(J_k, 0)}. \end{aligned} \quad (6.8)$$

Finally, the last case is

$$\text{dist}(J_n, J_k) < \min(|J_n|, |J_k|).$$

Again we assume that $n, k > 0$. Then by monotonicity either $n = k$ or $|n - k| = 1$, i.e. the intervals are either the same or adjacent. The estimates in this case are more complicated and will be done differently. First, integrating by parts we get

$$\begin{aligned} \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} &= \int_{J_n} w' \left[\int_{J_k} \frac{w(t) dt}{t - x} \right] \frac{dx}{x^2} = \\ &= - \int_{J_n} w' \left[\int_{J_k} \log |t - x| w'(t) dt \right] \frac{dx}{x^2}. \end{aligned}$$

Next we would like to conclude that, for $k \leq n$,

$$\begin{aligned} &- \int_{J_n} w' \left[\int_{J_k} \log |t - x| w'(t) dt \right] \frac{dx}{x^2} \lesssim \\ &- \frac{1}{1 + \text{dist}^2(J_n, 0)} \left[\iint_{J_n \times J_k} \log |t - x| w'(x) w'(t) dx dt + |J_n|^2 \right] \end{aligned} \quad (6.9)$$

and work with the latter integral instead of the former. Since $\text{dist}(0, J_n) < |x|$ for $x \in |J_n|$, this estimate would be obvious if the function under the integral were negative. In our case, however, it will require some work.

To prove the last relation denote

$$J_n^+ = \{w' > 0\}, \quad J_n^- = \{w' \leq 0\}.$$

Since $-\int_{J_k} \log |t - x| w'(t) dt = \tilde{w}_k(x)$ and w_k is an atom,

$$\left\| \int_{J_k} \log |t - x| w'(t) dt \right\|_{L^1_{\mathbb{H}}} \lesssim \frac{|J_k|^2}{1 + \text{dist}^2(J_k, 0)}.$$

Since $\{J_n\}$ is a short sequence, $|J_n| \leq \text{dist}(J_n, 0)$ for large enough n . For the rest of the proof we will assume that this holds for all $n \neq 0, -1$. Since w' is bounded from below and

$$\text{dist}^2(J_n, 0) \leq x^2 \leq 2\text{dist}^2(J_n, 0)$$

on J_n ,

$$\begin{aligned} & - \int_{J_n^-} w' \left[\int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{x^2} \lesssim \\ & - \int_{J_n^-} w' \left[\int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{1 + \text{dist}^2(J_n, 0)} + \frac{|J_k|^2}{1 + \text{dist}^2(J_k, 0)}. \end{aligned} \quad (6.10)$$

To deal with the integral over J_n^+ notice, that $\tilde{w}_k$ is "almost" positive on J_n , i.e.

$$- \int_{J_k} \log |t-x| w'(t) dt \gtrsim -|J_n| \quad (6.11)$$

for any $x \in J_n$. Indeed

$$- \int_{J_k} \log |t-x| w'(t) dt = \int_{J_k} \log_- |t-x| w'(t) dt - \int_{J_k} \log_+ |t-x| w'(t) dt,$$

where

$$\begin{aligned} \int_{J_k} \log_- |t-x| w'(t) dt &= \int_{J_k} \log_- |t-x| (\arg \theta)'(t) dt - \int_{J_k} \log_- |t-x| (x+v)'(t) dt \geq \\ & \int_{J_k} (\arg \theta)'(t) dt - \int_{J_k} (1+\varepsilon) dt \geq -\varepsilon |J_k| + \text{const}. \end{aligned}$$

Here we used the property that $\int_{J_k} (\arg \theta)'(t) dt = |J_k| + \text{const}$. Also

$$- \int_{J_k} \log_+ |t-x| w'(t) dt = \int_{J_k} \log_+ |t-x| (x+v)'(t) dt - \int_{J_k} \log_+ |t-x| (\arg \theta)'(t) dt.$$

Recall that $J_k = (b_k, b_{k+1}]$ and $x \in J_n$, $n \geq k$. If $n > k$, sing lemma 6 part 6 we obtain

$$- \int_{J_k} \log_+ |t-x| w'(t) dt \geq (\log_+ |b_k-x| - C) \int_{J_k} (1+v') dx - \log_+ |b_k-x| \int_{J_k} (\arg \theta)' dx \gtrsim -|J_k|,$$

which establishes (6.11). For $n = k$ the same relation can be obtained using lemma 6 part 5.

To finish the proof of (6.9) notice that

$$\begin{aligned} & - \int_{J_n} w' \left[\int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{x^2} = \\ & \int_{J_n^+} w' \left[- \int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{x^2} + \int_{J_n^-} w' \left[- \int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{x^2} = \\ & \int_{J_n^+} w'(x) \max \left(\left[- \int_{J_k} \log |t-x| w'(t) dt \right], 0 \right) \frac{dx}{x^2} + \\ & \int_{J_n^+} w'(x) \min \left(\left[- \int_{J_k} \log |t-x| w'(t) dt \right], 0 \right) \frac{dx}{x^2} + \\ & \int_{J_n^-} w' \left[- \int_{J_k} \log |t-x| w'(t) dt \right] \frac{dx}{x^2}. \end{aligned}$$

Regarding the last three integrals, if one replaces $\frac{dx}{x^2}$ with $\frac{dx}{1 + \text{dist}^2(J_n, 0)}$ in the first integral, it will get $\gtrsim$ than before because the function under the integral is positive and $x^2 \gtrsim 1 + \text{dist}^2(J_n, 0)$. Under the same operation, the second integral will decrease at most by

$$C |J_k| \int_{J_n} |w'| \frac{dx}{x^2} \lesssim \frac{|J_n| |J_k|}{1 + \text{dist}^2(J_n, 0)}$$

because of (6.11). Finally, the last integral is estimated in (6.10). Since $\text{dist}(0, J_k) \asymp \text{dist}(0, J_n)$ and $|J_k| \leq |J_n|$, this finishes (6.9).

To estimate the integral in the right-hand side of (6.9), denote $p = \arg \theta - x - v_1 = w + v_2$ where the functions v_1, v_2 are from claim 1. Also denote $p_n = p|_{J_n}$ and $v_2^n = v_2|_{J_n}$. The key properties of v_1 that we will use are that $\arg \theta - x - v_1 = 0$ at the endpoints of all I_n , $v_2 = 0$ at the endpoints of J_n and that $|v_1'|, |v_2'| < \varepsilon$. Then

$$\begin{aligned} - \iint_{J_n \times J_k} \log |t - x| w'(x) w'(t) dx dt &= - \iint_{J_n \times J_k} \log |t - x| p'(x) p'(t) dx dt - \\ &\quad + \int_{J_n} (\tilde{p}_n v_2' + \tilde{v}_2^n p' + \tilde{v}_2^n v_2') dx. \end{aligned}$$

Notice that

$$\left| \int_{J_n} \tilde{p}_n v_2' dx \right| \leq \varepsilon \|\tilde{p}_n\|_1 \leq \varepsilon \|p_n\|_2 \sqrt{|J_n|} \lesssim \varepsilon |J_n|^2$$

because $|p_n| \lesssim |J_n|$. Also,

$$\left| \int_{J_n} p' \tilde{v}_2^n dx \right| = | \langle p, v_2^n \rangle_{\mathcal{D}} | = \left| \int_{J_n} \tilde{p}_n v_2' dx \right| \lesssim |J_n|^2.$$

Similarly to the first integral,

$$\int_{J_n} \tilde{v}_2^n v_2' dx \lesssim \varepsilon^2 |J_n|^2.$$

Hence

$$- \iint_{J_n \times J_k} \log |t - x| w'(x) w'(t) dx dt = - \iint_{J_n \times J_k} \log |t - x| p'(x) p'(t) dx dt + O(|J_n|^2). \quad (6.12)$$

For the last integral we have

$$- \iint_{J_n \times J_k} \log |t - x| p'(x) p'(t) dx dt = - \sum_{I_i \subset J_k} \sum_{I_j \subset J_n} \iint_{I_i \times I_j} \log |t - x| p'(x) p'(t) dx dt. \quad (6.13)$$

To estimate

$$\begin{aligned} &- \iint_{I_i \times I_j} \log |t - x| p'(x) p'(t) = \\ &\quad \iint_{I_i \times I_j} \log_- |t - x| p'(x) p'(t) - \iint_{I_i \times I_j} \log_+ |t - x| p'(x) p'(t) \end{aligned} \quad (6.14)$$

we consider 3 cases. First, to estimate the integral in the case when $i = j$, notice that, since $1 + v_1'$ is bounded,

$$\int_{I_j} \log_- |x - t| (1 + v_1'(x)) dx < \text{const}$$

for any $t \in I_j$. Once again, the positive functions $\arg' \theta$ and $v_1' + 1$ satisfy

$$\int_{I_l} \arg' \theta = \int_{I_l} (v_1' + 1) = 2\pi \Delta_l + O(1) = |I_l| + O(1). \quad (6.15)$$

Hence

$$\begin{aligned} & \iint_{I_j \times I_j} \log_- |t-x| p'(x) p'(t) = \\ & \iint_{I_j \times I_j} \log_- |t-x| (\arg \theta)'(x) (\arg \theta)'(t) - 2 \iint_{I_j \times I_j} \log_- |t-x| (1+v'_1(x)) (\arg \theta)'(t) + \\ & \iint_{I_j \times I_j} \log_- |t-x| (1+v'_1(x)) (1+v'_1(t)) = \iint_{I_j \times I_j} \log_- |t-x| (\arg \theta)'(x) (\arg \theta)'(t) + O(|I_j|). \end{aligned}$$

For the last integral we have

$$\begin{aligned} & \iint_{I_j \times I_j} \log_- |t-x| (\arg \theta)'(x) (\arg \theta)'(t) = \\ & \sum_{\lambda_l, \lambda_{l+1} \subset I_j} \sum_{\lambda_m, \lambda_{m+1} \subset I_j} \int_{\lambda_l}^{\lambda_{l+1}} \int_{\lambda_m}^{\lambda_{m+1}} (\arg \theta)'(x) (\arg \theta)'(t). \end{aligned}$$

Using the properties that

$$\int_{\lambda_s}^{\lambda_{s+1}} \arg \theta' = 2\pi$$

and

$$\arg \theta' \lesssim [\min(|I_{s-1}|, |I_s|, |I_{s+1}|)]^{-2} \quad \text{on } (\lambda_s, \lambda_{s+1}),$$

for all s , we can apply lemma 6, parts 1-3. Assuming that $\lambda_l \leq \lambda_m$ we conclude that

$$\begin{aligned} & \int_{\lambda_l}^{\lambda_{l+1}} \int_{\lambda_m}^{\lambda_{m+1}} \log_- |t-x| (\arg \theta)'(x) (\arg \theta)'(t) \lesssim \\ & \begin{cases} \log_-(\lambda_m - \lambda_{l+1}) & \text{if } \lambda_m > \lambda_{l+1} \\ \max(\log_-(\lambda_{l-1} - \lambda_l), \log_-(\lambda_l - \lambda_{l+1}), \log_-(\lambda_{l+1} - \lambda_{l+2})) + 1 & \text{if } \lambda_m = \lambda_{l+1} \\ \max(\log_-(\lambda_{l-1} - \lambda_l), \log_-(\lambda_l - \lambda_{l+1}), \log_-(\lambda_{l+1} - \lambda_{l+2})) + 1 & \text{if } \lambda_m = \lambda_l, \end{cases} \end{aligned}$$

which implies

$$\begin{aligned} & \iint_{I_j \times I_j} \log_- |t-x| p'(x) p'(t) dx dt \lesssim \\ & \sum_{\lambda_{k_n} \leq \lambda_k, \lambda_l \leq \lambda_{k_{n+1}}, \lambda_k \neq \lambda_l} \log_- |\lambda_k - \lambda_l| + |I_j|. \end{aligned} \quad (6.16)$$

To estimate the integral of $\log_+$, first notice that by lemma 6, part 5, and (6.15),

$$\int_{I_j} \log_+ |x-t| (1+v'_1(x)) dx = |I_j| \log_+ |I_j| + O(|I_j|)$$

for any $t \in I_j$.

Together with part 4 of lemma 6 and (6.15) we get :

$$\begin{aligned} & \iint_{I_j \times I_j} \log_+ |t-x| p'(x) p'(t) = \iint_{I_j \times I_j} \log_+ |t-x| \arg' \theta(x) \arg' \theta(t) - \\ & 2 \iint_{I_j \times I_j} \log_+ |t-x| (v'_1(x) + 1) \arg' \theta(t) + \iint_{I_j \times I_j} \log_+ |t-x| (v'_1(x) + 1) (v'_1(t) + 1) = \end{aligned}$$

$$\begin{aligned}
& \sum_{\lambda_l, \lambda_m \in I_j} \int_{\lambda_l}^{\lambda_{l+1}} \int_{\lambda_m}^{\lambda_{m+1}} \log_+ |t-x| \arg' \theta(x) \arg' \theta(t) - |I_j|^2 \log |I_j| + O(|I_j|^2) \geq \\
& 4\pi^2 \sum_{\lambda_l, \lambda_m \in I_j} \log_+ |\lambda_l - \lambda_m| - |I_j|^2 \log |I_j| + O(|I_j|^2) \quad (6.17)
\end{aligned}$$

Next, let us consider the case when $i \neq j$ and the intervals I_i, I_j are not adjacent. This estimate is similar to (6.8), but we will do it using a different technique. Assume for instance that $j > i + 1$. For $\log_-$, recalling that $|I_k| > 1$ for all k , we get

$$- \iint_{I_i \times I_j} \log_- |t-x| p'(x) p'(t) dx dt = 0. \quad (6.18)$$

For $\log_+$ we have

$$\begin{aligned}
& - \iint_{I_i \times I_j} \log_+ |t-x| p'(x) p'(t) = - \int_{a_i}^{a_{i+1}} \int_{a_j}^{a_{j+1}} \log_+ |t-x| p'(x) p'(t) = \\
& - \int_{a_i}^{a_{i+1}} \int_{a_j}^{a_{j+1}} \log_+ |t-x| (\arg \theta - x - v_1)'(x) (\arg \theta - x - v_1)'(t) \leq \\
& - \int_{I_j} \left(\log |a_{i+1} - t| \int_{I_i} \arg' \theta(x) dx - \log |a_i - t| \int_{I_i} (v_1' + 1)(x) dx \right) \arg' \theta(t) dt + \\
& \int_{I_j} \left(\log |a_i - t| \int_{I_i} \arg' \theta(x) dx - \log |a_{i+1} - t| \int_{I_i} (v_1' + 1)(x) dx \right) (v_1' + 1)(t) dt \leq \\
& 2|I_i||I_j| \quad (6.19)
\end{aligned}$$

Here we used the properties that $\text{dist}(I_i, I_j) \geq |I_i|$ by monotonicity and (6.15).

In the case when I_i and I_j are adjacent, i.e. $j = i + 1$, the estimate can be done differently. Note that $p_n = p_n|_{I_n}$ is a compactly supported function with bounded derivative (the bound depends on n). Therefore it belongs to the Dirichlet space $\mathcal{D}$. The estimates (6.16) and (6.17) yield

$$\|p_n\|_{\mathcal{D}}^2 \lesssim \frac{1}{4\pi^2} |I_n|^2 \log |I_n| - E_n + |I_n|^2.$$

Hence

$$\begin{aligned}
& \iint_{I_i \times I_{i+1}} \log |t-x| p'(x) p'(t) = \langle p_i, p_{i+1} \rangle_{\mathcal{D}} \leq \|p_i\|^2 + \|p_{i+1}\|^2 \lesssim \\
& \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i \right) + \left(\frac{1}{4\pi^2} |I_{i+1}|^2 \log |I_{i+1}| - E_{i+1} \right) + \\
& |I_i|^2 + |I_{i+1}|^2. \quad (6.20)
\end{aligned}$$

Now we can return to estimating

$$\int_{J_n} w' \tilde{w}_k \frac{dx}{x^2}$$

in the case when $|k-n| \leq 1$. Using the estimates (6.9), (6.12) and (6.13) we obtain

$$\int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} = \frac{-\sum_{I_i \subset J_k} \sum_{I_j \subset J_n} \iint_{I_i \times I_j} \log |t-x| p'(x) p'(t) dx dt + O(|J_n|^2)}{1 + \text{dist}^2(0, J_n)}.$$

The estimates (6.16 – 6.20) yield

$$\begin{aligned} & - \sum_{I_i \subset J_n} \sum_{I_j \subset J_k} \iint_{I_i \times I_j} \log |t-x| p'(x) p'(t) dx dt \lesssim \\ & \sum_{I_i \subset J_k \cup J_n} \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i + |I_i|^2 \right) + \sum_{I_i, I_j \subset J_n} |I_i| |I_j| \leq \\ & \sum_{I_i \subset J_k \cup J_n} \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i \right) + |J_n|^2 + |J_k|^2. \end{aligned}$$

All in all, in the case $|n-k| \leq 1$, we have

$$\begin{aligned} & \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} \lesssim \\ & \frac{1}{1 + \text{dist}^2(0, J_n)} \left[\sum_{I_i \subset J_k \cup J_n} \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i \right) + |J_n|^2 + |J_k|^2 \right]. \end{aligned} \quad (6.21)$$

Combining the estimates (6.16), (6.17), (6.18), (6.19), (6.20) with (6.13), (6.12) and (6.9) we get

$$\begin{aligned} & \sum_n \sum_k \int_{J_n} w' \tilde{w}_k \frac{dx}{x^2} = \\ & \sum_n \left[\sum_{k : \max(|J_k|, |J_n|) \leq \text{dist}(J_k, J_n)} \right] + \\ & \sum_n \left[\sum_{k : \min(|J_k|, |J_n|) \leq \text{dist}(J_k, J_n) < \max(|J_k|, |J_n|)} \right] + \\ & \sum_n \left[\sum_{k : \text{dist}(J_k, J_n) < \min(|J_k|, |J_n|)} \right] = \\ & I + II + III. \end{aligned}$$

For the first sum, by (6.6), we get

$$\begin{aligned} I & \lesssim \sum_k \left[\sum_{n : n > k, \text{dist}(J_k, J_n) > |J_k|} \frac{|J_k|^3}{1 + \text{dist}^2(J_n, 0)} \int_{J_n} \frac{dx}{\text{dist}^2(J_k, x)} \right] \leq \\ & \sum_k \frac{|J_k|^3}{1 + \text{dist}^2(J_k, 0)} \frac{1}{|J_k|} = \sum_k \frac{|J_k|^2}{1 + \text{dist}^2(J_k, 0)} < \infty. \end{aligned} \quad (6.22)$$

For the second sum, by (6.8),

$$II \lesssim \sum_n \left[\sum_{k : k \neq n, \text{dist}(J_k, J_n) < \max(|J_k|, |J_n|)} \frac{|J_k||J_n|}{1 + \text{dist}^2(J_n, 0)} \right].$$

Recall that by our assumption $|J_n| \leq \text{dist}(J_n, 0)$ for all $n \neq 0, -1$. We can also assume that $|J_{-1}| = |J_0|$. Then in each term in the last sum k and n have the same sign. Let us estimate the part of the sum with non-negative k, n .

$$\begin{aligned} & \sum_{n \geq 0} \left[\sum_{k \geq 0 : k \neq n, \text{dist}(J_k, J_n) < \max(|J_k|, |J_n|)} \frac{|J_k||J_n|}{1 + \text{dist}^2(J_n, 0)} \right] = \\ & 2 \sum_{n \geq 0} \left[\sum_{k \geq 0 : k < n, \text{dist}(J_k, J_n) < |J_n|} \frac{|J_k||J_n|}{1 + \text{dist}^2(J_n, 0)} \right] \leq 4 \sum_{n \geq 0} \frac{|J_n|^2}{1 + \text{dist}^2(J_n, 0)} < \infty. \end{aligned}$$

The negative terms can be estimated similarly to conclude that

$$II \lesssim \sum_n \frac{|J_n|^2}{1 + \text{dist}^2(J_n, 0)} < \infty.$$

Finally, for the third sum by (6.21),

$$\begin{aligned} III & \lesssim \sum_n \left[\sum_{k : |k-n| \leq 1} \frac{1}{1 + \text{dist}^2(0, J_n)} \left[\sum_{I_i \subset J_k \cup J_n} \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i \right) + |J_n|^2 + |J_k|^2 \right] \right] \lesssim \\ & \sum_n \frac{1}{1 + \text{dist}^2(0, J_n)} \left[\sum_{I_i \subset J_n} \left(\frac{1}{4\pi^2} |I_i|^2 \log |I_i| - E_i \right) + |J_n|^2 \right] < \infty \end{aligned}$$

Because Λ satisfies the energy condition on I_n . Altogether these estimates give us

$$\int_{\mathbb{R}} \tilde{w} w' \frac{dx}{x^2} < \infty.$$

The integrals over the circular part of $\partial D(r)$ can be estimated like in [20], section 5.3. We need to show that the integrals

$$\int_{\partial D(r) \setminus \mathbb{R}} q \, d\tilde{q} = rI'(r), \quad I(r) := \frac{1}{2} \int_0^\pi q^2 (re^{i\phi}) \, d\phi,$$

do not tend to $+\infty$ as $r \rightarrow \infty$. In fact, it is enough to show

$$I(r) \not\rightarrow \infty,$$

because if $rI'(r) \rightarrow +\infty$, then $I'(r) \geq 1/r$ for all $r \gg 1$, and we have $I(r) \rightarrow \infty$.

As we will see shortly, $I(r) \not\rightarrow \infty$ for *any* $q \in L^1(1 + |x|^{-1})$. It will be more convenient for us to prove an equivalent statement in the unit disk $\mathbb{D}$.

Let $q + i\tilde{q}$ be an analytic function in $\mathbb{D}$ such that

$$\frac{q(\zeta)}{1 - |\zeta|} \in L^1(\partial \mathbb{D}).$$

Define

$$h(z) = \frac{1+z}{1-z} (q(z) + i\tilde{q}(z)), \quad z \in \mathbb{D},$$

and denote by $h^M(\zeta)$, $\zeta \in \partial\mathbb{D}$, the angular maximal function. Then $\Im h \in L^1(\partial\mathbb{D})$ and by the Hardy-Littlewood maximal theorem,

$$h^M \in L^{1,\infty}(\partial\mathbb{D}). \quad (6.23)$$

Let us show that as $\epsilon \rightarrow 0$,

$$\frac{1}{\epsilon} \int_{C_\epsilon} |q + i\tilde{q}|^2 |dz| \not\rightarrow \infty, \quad C_\epsilon := \{|1 - z| = \epsilon\} \cap \mathbb{D}.$$

We have

$$\frac{1}{\epsilon} \int_{C_\epsilon} |q + i\tilde{q}|^2 \leq \epsilon \int_{C_\epsilon} |h|^2 \lesssim [\epsilon h^M(\zeta)]^2 + [\epsilon h^M(\bar{\zeta})]^2,$$

where $\zeta \in \partial\mathbb{D}$, $|1 - \zeta| = \epsilon$. The right-hand side cannot tend to infinity because otherwise, for all small ϵ , we would have

$$h^M(\zeta) + h^M(\bar{\zeta}) \gg \frac{1}{\epsilon}$$

on an interval of length ϵ , which would contradict (6.23). $\triangle$

Let

$$\phi = \arg(\theta \bar{S} I_+ \bar{I}_-)/2.$$

Recall that $\tilde{\phi} \in L^1_\Pi$. By the last claim ϕ/x belongs to the Dirichlet class. Hence, by the Beurling-Malliavin multiplier theorem, see for instance [20], there exists a smooth function m on $\mathbb{R}$ satisfying

$$m' < \varepsilon, \quad \tilde{m} \in L^1_\Pi \quad \text{and} \quad \tilde{m} \geq \max(0, -\tilde{\phi}).$$

In other words, if Φ and M are outer functions,

$$\Phi = \exp(i\phi - \tilde{\phi}), \quad M = \exp(im - \tilde{m}),$$

then ΦM is bounded in $\mathbb{C}_+$.

Since $m' < \varepsilon$, $\varepsilon x - m$ is an increasing function. There exists a meromorphic inner function J such that

$$\{J = \pm 1\} = \{2(\varepsilon x - m) = k\pi\}.$$

Denote

$$d_1 = 2(\varepsilon x - m) \quad \text{and} \quad d_2 = \arg J.$$

Then the difference

$$d = 2(\varepsilon x - m) - \arg J = d_1 - d_2$$

satisfies $|d| < \pi$.

Put

$$l(x) = \frac{2\varepsilon x - \arg J}{2}.$$

Notice that $\tilde{l} \in L^1_\Pi$ because $2l = d + 2m$ where d is bounded and $\tilde{m} \in L^1_\Pi$.

Consider an outer function $\Psi = \exp(il - \tilde{l})$. Then

$$\bar{S}^{2\varepsilon} \Psi = \bar{J} \bar{\Psi}$$

or equivalently

$$\bar{S}^{2\varepsilon} J \Psi = \bar{\Psi}$$

on $\mathbb{R}$. Thus $\Psi \in N^+[\bar{S}^{2\varepsilon} J]$.

Moreover, the ratio Ψ/M is equal to $\exp\left(i\frac{d}{2} - \frac{\tilde{d}}{2}\right)$. Since $|d| < \pi$, Ψ/M belongs to any L_{Π}^p , $p < 1$. Our next goal is to construct another "small" outer multiplier function k so that $k\Psi/M \in L_{\Pi}^2$.

Consider the step function

$$\alpha(x) = \frac{\pi}{5} \left[\frac{5}{\pi} d_1 \right] - \frac{\pi}{5} \left[\frac{5}{\pi} d_2 \right],$$

where $[\cdot]$ again denotes the entire part of a real number. Then

$$|d - \alpha| < \frac{2\pi}{5}. \quad (6.24)$$

Since $d_1 = d_2 = \pi n$ at the points $\{c_n\} = \{J = \pm 1\}$, the function α only takes values $k\frac{\pi}{5}$, $k = -4, -3, \dots, 4$. Therefore α can be represented as

$$\alpha = \frac{\pi}{5} \left(\sum_{n=1}^4 \beta_n - \sum_{n=5}^8 \beta_n \right),$$

where β_n are elementary step functions, each taking only two values, 0 and 1, and making at most one positive and one negative jump on each interval $[c_n, c_{n+1}]$. For each $n = 1, 2, \dots, 8$ one can choose an inner function Q_n so that

$$\frac{1 - Q_n}{1 + Q_n} = \text{const } e^{\pi K \beta_n}.$$

Notice that then

$$\exp(-i\pi\alpha + \pi\tilde{\alpha}) = \text{const} \sqrt[5]{\prod_{n=1}^4 \frac{1 + Q_n}{1 - Q_n} \prod_{n=5}^8 \frac{1 - Q_n}{1 + Q_n}}.$$

Because of (6.24) we have

$$\begin{aligned} \left| \prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \Psi/M \right| &\lesssim \left| \sqrt[10]{\prod_{n=1}^4 \frac{1 + Q_n}{1 - Q_n} \prod_{n=5}^8 \frac{1 - Q_n}{1 + Q_n}} \Psi/M \right| \\ &= \text{const} \exp \left[\frac{\tilde{d}}{2} - \frac{\tilde{\alpha}}{2} \right] \in L_{\Pi}^2(\mathbb{R}) \end{aligned}$$

and since the function $M\Phi$ is bounded,

$$\prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \Psi\Phi = \prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \frac{\Psi}{M} M\Phi \in L_{\Pi}^2(\mathbb{R}).$$

Now notice that since $N^+[\bar{S}^{2\varepsilon} J] \neq 0$, the set $\{c_n\} = \{J = \pm 1\}$ has Beurling-Malliavin density at most 2ε , see section 7 or [19]. By our construction the Beurling-Malliavin density of each of the sets $\{Q_n = \pm 1\}$ is the same as that of $\{c_n\}$, i.e. at most 2ε . Hence the kernel $N^\infty[\bar{S}^{17\varepsilon} \prod_n Q_n]$ contains a non-zero function τ , see section 7 or [19].

Similarly, since the Beurling-Malliavin density of $\{I_+ = 1\}$ is less than ε , the kernel $N^\infty[\bar{S}^\varepsilon I_+]$ is infinite-dimensional. Hence it contains a non-trivial function η

with at least one zero a in $\mathbb{C}_+$. Then the function $\kappa = \eta/(z - a)$ also belongs to $N^\infty[\bar{S}^\varepsilon I_+]$ and satisfies $|\kappa| \lesssim (1 + |x|)^{-1}$ on $\mathbb{R}$.

Therefore

$$\begin{aligned} & \bar{\theta} S^{1-20\varepsilon} \kappa \tau \prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \Psi \Phi = \\ & (\bar{S}^\varepsilon I_+ \kappa) \left(\bar{S}^{17\varepsilon} \prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \tau \right) (\bar{S}^{2\varepsilon} \Psi) (\bar{\theta} S^1 \bar{I}_+ \Phi) \in \bar{H}^2. \end{aligned}$$

Hence the space K_θ contains the function

$$f = S^{1-20\varepsilon} \kappa \tau \prod_{n=1}^4 (1 + Q_n) \prod_{n=5}^8 (1 - Q_n) \Psi \Phi.$$

Now we could simply refer to theorem 1 to conclude this part of the proof. For the convenience of those readers who skipped theorem 1, we also present a direct argument.

By the Clark representation formula

$$f = (1 + \theta) K f \sigma_1$$

where σ_1 is the Clark measure corresponding to θ concentrated on $\{\theta = 1\} = \Lambda$. Since $1 + \theta$ is bounded in the upper half-plane and f decreases faster than $\exp[-(1 - 21\varepsilon)y]$ along the positive y -axis, so does $K f \mu$. Hence $f \mu$ is the measure concentrated on Λ with the spectral gap at least $(1 - 21\varepsilon)$.

II) Now suppose that $G_X > 1$ but $C_X < \frac{1}{2\pi}$.

By corollary 4 there exists a discrete increasing sequence $\Lambda = \{\lambda_n\}_{n \in \mathbb{Z}} \subset X$ and a measure ν , $\text{supp } \nu = \Lambda$ such that ν has a spectral gap of the size 1 and $K\nu$ does not have any zeros in $\mathbb{C}$.

Similarly to the previous part, we assume that $\sup_n (\lambda_n - \lambda_{n-1}) < \infty$. The general case is discussed at the end of the proof. If $\sup_n (\lambda_n - \lambda_{n-1}) < \infty$, we can apply lemma 5 and consider the inner function θ corresponding to Λ . A function $f \in N[\phi]$ is called purely outer if f is outer in the upper half-plane and $\phi f = \bar{g}$ is outer in the lower half plane. Since $K\nu$ is divisible by S , the function

$$f = S^{-1}(1 - \theta) K \nu \in K_\theta^{1,\infty}$$

is a purely outer element of $N^{1,\infty}[\bar{\theta} S]$, see section 8. Note that $f = \exp(i\phi - \tilde{\phi})$ in $\mathbb{C}_+$, where $2\phi = \arg \theta - x$.

Denote by Γ_n the middle one-third of the interval $(\lambda_n, \lambda_{n+1})$. Our plan is to calculate the integral

$$\int_{\cup \Gamma_n} \phi' \tilde{\phi} \frac{dx}{x^2} \quad (6.25)$$

in two different ways and arrive at a contradiction by obtaining two different answers.

First let us choose a short monotone partition $\{I_n\}$ of $\mathbb{R}$ such that Λ satisfies the density condition (5.3) with $a = \frac{1}{2\pi}$ on that partition:

Put $a_0 = 0$. Choose $a_1 > a_0$ to be the smallest point such that $\#\Lambda \cap (a_0, a_1] \geq \frac{1}{2\pi}(a_1 - a_0)$. Note that such a point always exists because Λ supports a measure with

a spectral gap greater than 1: otherwise we would be able to choose a long sequence of intervals satisfying (7.1) in lemma 2 with $a = \frac{1}{2\pi}$ and arrive at a contradiction. After a_i , $i \geq 1$ is chosen, choose $a_{i+1} > a_i$ as the smallest point such that

$$\#\Lambda \cap (a_i, a_{i+1}] \geq \frac{1}{2\pi}(a_{i+1} - a_i) \quad \text{and} \quad (a_{i+1} - a_i) \geq (a_i - a_{i-1}).$$

Choose $a_i, i < 0$ in a similar way. Put $I_n = (a_n, a_{n+1}]$. Again by lemma 2, $\{I_n\}$ has to be short.

In what follows we will assume, WLOG, that $\frac{1}{2\pi}|I_n| = \#\Lambda \cap I_n$.

Note that since $C_X < 1$, the sum in the energy condition (5.4) has to be infinite. At the same time, a part of that sum has to be finite:

Claim 3.

$$\sum_n \frac{\log_-(\lambda_{n+1} - \lambda_n)}{\lambda_n^2} < \infty.$$

Proof of claim. Suppose that the sum is infinite. Put $\mu = |\nu|$ and let Φ be the inner function corresponding to μ . Let $\psi = \arg \Phi - x$.

Define the intervals J_n and the function v like in part I) of the proof, with Φ replacing θ . Put $w = \psi - v = \arg \Phi - x - v$. Let again $w_n = |J_n|$, $w_n^* = w - w_n$. Then $\tilde{w} \in L^1_\Pi$ because w_n are atoms with summable L^1_Π -norms.

Like in part I) we can use "atomic" estimates to show that if $\text{dist}(J_k, J_n) > \max(|J_k|, |J_n|)$ and $x \in J_n$ then

$$|\tilde{w}_k(x)| \lesssim \frac{|J_k|^3}{\text{dist}^2(x, J_k)}.$$

By monotonicity and shortness of J_k we conclude that

$$\sum_{\lambda_i \in J_n} \frac{|\tilde{w}_k(\lambda_i)|}{\lambda_i^2} \lesssim \sum_{\lambda_i \in J_n} \frac{|\tilde{w}_k(\lambda_i)|}{1 + \text{dist}^2(0, J_n)} \lesssim \frac{1}{1 + \text{dist}^2(0, J_n)} \int_{J_n} \frac{|J_k|^3}{\text{dist}^2(x, J_k)} dx.$$

Hence, similarly to (6.22),

$$\begin{aligned} & \sum_n \left[\sum_{k : \text{dist}(J_k, J_n) > \max(|J_k|, |J_n|)} \left[\sum_{\lambda_i \in J_n} \frac{|\tilde{w}_k(\lambda_i)|}{\lambda_i^2} \right] \right] = \\ & 2 \sum_n \left[\sum_{k : k < n, \text{dist}(J_k, J_n) > \max(|J_k|, |J_n|)} \left[\sum_{\lambda_i \in J_n} \frac{|\tilde{w}_k(\lambda_i)|}{\lambda_i^2} \right] \right] \lesssim \\ & \sum_n \left[\sum_{k : k < n, \text{dist}(J_k, J_n) > \max(|J_k|, |J_n|)} \frac{1}{1 + \text{dist}^2(0, J_n)} \int_{J_n} \frac{|J_k|^3}{\text{dist}^2(x, J_k)} dx \right] < \infty. \end{aligned}$$

In other words, on each J_n

$$\sum_{k : \text{dist}(J_k, J_n) > \max(|J_k|, |J_n|)} |\tilde{w}_k^*| \leq g_1$$

where g_1 is a positive function satisfying

$$\sum_n \frac{g_1(\lambda_n)}{1 + \lambda_n^2} < \infty.$$

Also for any $x \in J_n$

$$\tilde{w}_k(x) = \int \frac{dt}{t-x} w_k(t) = - \int_{J_k} \log |t-x| w'(t) dt.$$

If $k < n$ then

$$\begin{aligned} & - \int_{J_k} \log |t-x| w'(t) dt \geq \\ & - \int_{J_k} \log_+ |t-x| (\arg \Phi)'(t) dt + \int_{J_k} \log_+ |t-x| (1+v')(t) dt - \text{const} \gtrsim -|J_k|. \end{aligned} \quad (6.26)$$

Here we applied lemma 6, part 6, to the second integral in the second line and used the estimate

$$- \int_{J_k} \log_+ |t-x| (\arg \Phi)'(t) dt \geq -\log(x-b) \int_{J_k} (\arg \Phi)',$$

where b is the left endpoint of J_k , for the first integral.

Thus for $x \in J_n$

$$\sum_{k : k \neq n, \text{ dist}(J_k, J_n) < \max(|J_k|, |J_n|)} \tilde{w}_k(x) \geq g_2(x),$$

where again

$$\sum_n \frac{|g_2(\lambda_n)|}{1 + \lambda_n^2} < \infty.$$

Also, for $x \in J_n$

$$\begin{aligned} & - \int_{J_n} \log_+ |x-t| w'(t) dt = \\ & - \int_{J_n} \log_+ |x-t| \arg' \Phi(t) dt + \int_{J_n} \log_+ |x-t| (1-v')(t) dt - \text{const} \gtrsim -|J_n|. \end{aligned} \quad (6.27)$$

Here we again used the property that $\int_{J_n} \arg' \Phi = \int_{J_n} (v' - 1) = |J_n| + O(1)$ and applied lemma 6, part 5, to the second integral in the second line.

Hence for any $x \in \mathbb{R}$

$$\tilde{w}(x) \geq \int \log_- |x-t| w'(t) dt + g(x)$$

for some function g satisfying

$$\sum_n \frac{|g(\lambda_n)|}{1 + \lambda_n^2} < \infty.$$

Therefore,

$$\begin{aligned} & \sum_n \frac{\tilde{w}(\lambda_n)}{1 + \lambda_n^2} \geq \text{const} + \sum_n \frac{\int_{\lambda_{n-1}}^{\lambda_n+1} \log_- |\lambda_n - x| w' dx}{1 + \lambda_n^2} \geq \\ & \text{const} + \sum_n \frac{\int_{\lambda_{n-1}}^{\lambda_n} \log_- |\lambda_n - x| (\arg \Phi)' dx}{1 + \lambda_n^2} \geq \text{const} + \sum_n \frac{\log_- |\lambda_n - \lambda_{n-1}|}{1 + \lambda_n^2}. \end{aligned}$$

Let $f = (1 + \Phi)K\nu$. Then f is an outer function in $\mathbb{C}_+$ that belongs to H^2 and satisfies

$$f = \exp \left(i \frac{\psi}{2} - \frac{\tilde{\psi}}{2} \right).$$

Since $|f(\lambda_n)| = |\nu(\{\lambda_n\})|/\mu(\{\lambda_n\}) = 1$, we have that $\log |f(\lambda_n)| = 2\tilde{\psi}(\lambda_n) = 0$ for all n .

Recall that $\tilde{w}(\lambda_n) = \tilde{\psi}(\lambda_n) + \tilde{v}(\lambda_n) = \tilde{v}(\lambda_n)$. It is left to show that

$$\sum_n \frac{\tilde{v}(\lambda_n)}{1 + \lambda_n^2} < \infty.$$

Recall that $v \in L_{\Pi}^1$, $\tilde{v} = \tilde{w} - \tilde{\psi} = \tilde{w} - \log |f|/2 \in L_{\Pi}^1$ and v' is bounded on $\mathbb{R}$. Therefore the harmonic extension of v into $\mathbb{C}_+$ has a bounded x -derivative in $\mathbb{C}_+$. Hence $\tilde{v}_y$ is bounded in $\mathbb{C}_+$ as well.

On each interval J_n choose λ_{k_n} so that $v(\lambda_{k_n}) = \max_{\lambda_i \in J_n} v(\lambda_i)$. If the last sum is positive infinite then so is

$$\sum_n |J_n| \frac{\tilde{v}(\lambda_{k_n})}{1 + \text{dist}^2(0, J_n)}.$$

Because of the boundedness of $\tilde{v}_y$, $\tilde{v}(\lambda_{k_n} + i|J_n|) \geq \tilde{v}(\lambda_{k_n}) - C|J_n|$ and therefore

$$\sum_n |J_n| \frac{\tilde{v}(\lambda_{k_n} + i|J_n|)}{1 + \text{dist}^2(0, J_n)} = \infty.$$

Denote by $(\tilde{v})^M$ the maximal non-tangential function of $\tilde{v}$ in $\mathbb{C}_+$. The last equation implies that $(\tilde{v})^M \notin L_{\Pi}^1$. But that contradicts the property that both $\tilde{v}$ and v belong to L_{Π}^1 . $\triangle$

Now notice that if $x \in \Gamma_n$ then

$$|f(x)| = |(1 - \theta(x))K\nu(x)| \leq 2 \left| \int \frac{1}{t - x} d\nu(t) \right| \leq 6\|\nu\| |\lambda_{n+1} - \lambda_n|^{-1}. \quad (6.28)$$

Hence

$$\begin{aligned} \int_{\cup \Gamma_n} \phi' \tilde{\phi} \frac{dx}{x^2} &\lesssim \sum_n \frac{1}{\lambda_n^2} \int_{\Gamma_n} \arg' \theta \log_+ |f| dx + \text{const} \lesssim \\ \sum_n \frac{1}{\lambda_n^2} \int_{\Gamma_n} \arg' \theta dx \log_- |\lambda_{n+1} - \lambda_n| + \text{const} &\lesssim \\ \sum_n \frac{1}{\lambda_n^2} \log_- |\lambda_{n+1} - \lambda_n| + \text{const} &< \infty. \end{aligned} \quad (6.29)$$

It follows that

$$\begin{aligned} \int_{\cup \Gamma_n} w' \tilde{w} \frac{dx}{x^2} &= \int_{\cup \Gamma_n} \phi' \tilde{\phi} \frac{dx}{x^2} - \int_{\cup \Gamma_n} \phi' \tilde{v} \frac{dx}{x^2} - \\ \int_{\cup \Gamma_n} v' \tilde{\phi} \frac{dx}{x^2} + \int_{\cup \Gamma_n} v' \tilde{v} \frac{dx}{x^2} &< \infty. \end{aligned} \quad (6.30)$$

Indeed, arguing like at the end of the proof of the last claim, from the property that $(\tilde{v})^M \in L_{\Pi}^1$ we deduce that

$$\sum_n |J_n| \frac{\sup_{x \in J_n} |\tilde{v}(x)|}{1 + \text{dist}^2(0, J_n)} < \infty.$$

Therefore

$$\left| \int_{\cup \Gamma_k \cap J_n} \phi' \tilde{v} \frac{dx}{x^2} \right| \leq \int_{\cup \Gamma_k \cap J_n} |\phi'| dx \frac{\sup_{x \in J_n} |\tilde{v}(x)|}{1 + \text{dist}^2(0, J_n)} \asymp |J_n| \frac{\sup_{x \in J_n} |\tilde{v}(x)|}{1 + \text{dist}^2(0, J_n)}$$

and summing over all n we get

$$\left| \int_{\cup \Gamma_k} \phi' \tilde{v} \frac{dx}{x^2} \right| < \infty.$$

The second integral on the right-hand side of (6.30) is finite because v' is bounded and $\tilde{\phi} = \log |f|$ is in L_{Π}^1 . The last integral is finite because v' is bounded and $\tilde{v} = \tilde{\phi} - \tilde{w} \in L_{\Pi}^1$.

Denote

$$L_n = \cup_{\text{dist}(J_k, J_n) < \max(J_k, |J_n|)} J_k, \quad q_n = w|_{L_n} \quad \text{and} \quad q_n^* = w - q_n.$$

Then

$$\int_{\cup \Gamma_k} w' \tilde{w} \frac{dx}{x^2} = \sum_n \left[\int_{\cup \Gamma_k \cap J_n} w' \tilde{q}_n^* \frac{dx}{x^2} + \int_{\cup \Gamma_k \cap J_n} w' \tilde{q}_n \frac{dx}{x^2} \right].$$

The first integral can be, once again, estimated like in (6.6), i.e. using the property that each w_i is an atom, and the sum of such integrals shown to be finite. For the second integral, applying similar arguments that were used in the first part to prove (6.9) we obtain

$$\begin{aligned} & \int_{\cup \Gamma_k \cap J_n} w' \tilde{q}_n \frac{dx}{x^2} \gtrsim \\ & - \frac{1}{1 + \text{dist}^2(J_n, 0)} \int_{\cup \Gamma_k \cap J_n} w'(x) \left[\sum_{J_l \subset L_n} \int_{J_l} \log |x - t| w'(t) dt - |J_n| |J_l| \right] dx. \end{aligned} \quad (6.31)$$

Furthermore, because of (6.26),

$$\begin{aligned} & - \int_{\cup \Gamma_k \cap J_n} w'(x) \left[\sum_{J_l \subset L_n} \int_{J_l} \log |x - t| w'(t) dt \right] dx \gtrsim \\ & - \int_{\cup \Gamma_k \cap J_n} w'(x) \left[\int_{J_n} \log |x - t| w'(t) dt \right] dx - \sum_{J_l \subset L_n} |J_l| |J_n|. \end{aligned}$$

Let us remark right away that

$$\begin{aligned} & \sum_n \frac{1}{1 + \text{dist}^2(J_n, 0)} \sum_{J_l \subset L_n} |J_l| |J_n| \lesssim \\ & \sum_{l < n, \text{dist}(J_l, J_n) < \max(J_l, |J_n|)} \frac{|J_l| |J_n|}{1 + \text{dist}^2(J_n, 0)} \lesssim \sum_n \frac{|J_n|}{1 + \text{dist}^2(J_n, 0)} < \infty. \end{aligned}$$

To continue the estimates let us split the last integral:

$$\begin{aligned}
& - \int_{\cup \Gamma_k \cap J_n} w'(x) \left[\int_{J_n} \log |x - t| w'(t) dt \right] dx = \\
& \int_{\cup \Gamma_k \cap J_n} (\arg \theta)'(x) \left[\int_{J_n} \log |x - t| (v'(t) + 1)(t) dt \right] dx \\
& - \int_{\cup \Gamma_k \cap J_n} (\arg \theta)'(x) \left[\int_{J_n} \log |x - t| (\arg \theta)'(t)(t) dt \right] dx \\
& - \int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) \left[\int_{J_n} \log |x - t| (v'(t) + 1) dt \right] dx \\
& + \int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) \left[\int_{J_n} \log |x - t| (\arg \theta)'(t)(t) dt \right] dx = \\
& \quad I + II + III + IV.
\end{aligned}$$

To estimate *III* and *IV* denote by C the constant satisfying

$$\int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) dx = C|J_n|.$$

Notice that because $1 - 2\varepsilon < v' + 1 < 1 + 2\varepsilon$ and $\int_{J_n} v' + 1 = \int_{J_n} (\arg \theta)' = |J_n|$, for any $y \in J_n$,

$$\int_{J_n} \log |y - t| (v'(t) + 1) dt = |J_n| \log |J_n| + O(|J_n|) \quad (6.32)$$

and

$$III = - \int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) \left[\int_{J_n} \log |x - t| (v'(t) + 1) dt \right] dx = -C|J_n|^2 \log |J_n| + O(|J_n|^2).$$

To estimate *IV* observe that for any $t \in J_n$, if $\text{dist}(t, (\lambda_k, \lambda_{k+1})) \geq 1$ then

$$\begin{aligned}
& \int_{\Gamma_k} (v'(x) + 1) \log_+ |x - t| dx \geq \\
& \int_{\Gamma_k} (v'(x) + 1) dx \frac{\int_{\lambda_k}^{\lambda_{k+1}} \log_+ |x - t| dx}{\lambda_{k+1} - \lambda_k} - (\lambda_{k+1} - \lambda_k) \log 3
\end{aligned}$$

(recall that Γ_k is the middle one-third of $(\lambda_k, \lambda_{k+1})$). Consider a positive step function $\alpha(x)$ defined on each $(\lambda_k, \lambda_{k+1})$ as

$$\frac{\int_{\Gamma_k} (v'(x) + 1) dx}{\lambda_{k+1} - \lambda_k}.$$

Then $|\alpha - 1| \leq \varepsilon$ on J_n . Hence one can apply lemma 6 part 5 to conclude that, for any $t \in J_n$,

$$\begin{aligned}
& \int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) \log_+ |x - t| dx \geq \\
& \int_{J_n} \alpha(x) \log_+ |x - t| dx - \text{const}|J_n| \geq \left(\int_{J_n} \alpha(x) \right) \log |J_n| - \text{const}|J_n| = \\
& C|J_n| \log |J_n| - \text{const}|J_n|.
\end{aligned}$$

Therefore

$$IV = \int_{\cup \Gamma_k \cap J_n} (v'(x) + 1) \left[\int_{J_n} \log |x - t| (\arg \theta)'(t) dt \right] dx \geq$$

$$\left(\int_{J_n} (\arg \theta)' \right) (C|J_n| \log |J_n| - \text{const}|J_n|) = C|J_n|^2 \log |J_n| - \text{const}|J_n|^2.$$

Combining the estimates we get

$$III + IV \gtrsim -|J_n|^2.$$

To estimate II notice that

$$II = - \sum_{\Gamma_k \subset J_n} \int_{\Gamma_k} (\arg \theta)'(x) dx \sum_{\lambda_j, \lambda_{j+1} \in J_n} \int_{\lambda_j}^{\lambda_{j+1}} \log |x - t| (\arg \theta)'(t) dt.$$

If $t \in (\lambda_j, \lambda_{j+1})$ and $x \in \Gamma_k$ then

$$\log |x - t| \geq \begin{cases} \log |\lambda_j - \lambda_{k+1}| & \text{if } j < k \\ \log |\lambda_k - \lambda_{j+1}| & \text{if } j > k \\ \log |\lambda_{j+1} - \lambda_j| & \text{if } j = k \end{cases}.$$

Put $\alpha_k = \int_{\Gamma_k} (\arg \theta)'$. Then

$$II \geq - \sum_{\Gamma_k \subset J_n} \alpha_k \sum_{\lambda_j \in J_n, j \neq k} 2\pi \log |\lambda_k - \lambda_j| + A_n$$

where the constants A_n satisfy

$$\sum_n \frac{|A_n|}{1 + \text{dist}^2(0, J_n)} < \infty.$$

Using (6.32), I can be rewritten as

$$I = \sum_{\Gamma_k \subset J_n} \int_{\Gamma_k} (\arg \theta)'(x) \left[\int_{J_n} \log |x - t| (v'(t) + 1)(t) dt \right] dx =$$

$$\left(\sum_{\Gamma_k \subset J_n} \alpha_k \right) |J_n| \log |J_n| + B_n$$

where again

$$\sum_n \frac{|B_n|}{1 + \text{dist}^2(0, J_n)} < \infty.$$

By the left-hand side of the inequality in part 2) of lemma 5,

$$\alpha_k = \int_{\Gamma_k} (\arg \theta)' > c > 0$$

for all k . Therefore, since there are $\frac{1}{2\pi}|J_n|$ intervals Γ_k in J_n ,

$$I + II = \left(\sum_{\Gamma_k \subset J_n} \alpha_k \right) |J_n| \log |J_n| -$$

$$\sum_{\Gamma_k \subset J_n} \alpha_k \sum_{\lambda_j \in J_n, j \neq k} 2\pi \log |\lambda_k - \lambda_j| + A_n + B_n \gtrsim$$

$$\frac{1}{2\pi}|J_n|^2 \log |J_n| - \sum_{\lambda_j, \lambda_k \in J_n} 2\pi \log |\lambda_k - \lambda_j| + A_n + B_n.$$

Now, going back to (6.31), we obtain

$$\sum_n \int_{\cup \Gamma_k \cap J_n} w' \tilde{w}_n \frac{dx}{x^2} \gtrsim \sum_n \frac{\frac{1}{4\pi^2}|J_n|^2 \log |J_n| - \sum_{\lambda_j, \lambda_k \in J_n} \log |\lambda_k - \lambda_j| - |J_n|^2 + A_n + B_n}{1 + \text{dist}^2(J_n, 0)} + \text{const.}$$

It remains to notice that the sum on right-hand side is positive infinite because otherwise Λ would satisfy the energy condition (5.4) and C_X would be at least 1. This contradicts (6.30).

It remains to discuss the case when $\sup_n(\lambda_n - \lambda_{n-1}) = \infty$. If Λ is such a sequence, choose a large constant C and consider the set of all gaps R_k of Λ of the size larger than C :

$$R_k = (\lambda_{n_k}, \lambda_{n_k+1}), \quad \lambda_{n_k+1} - \lambda_{n_k} > C.$$

After that one can add a separated set of points in every R_k and consider a slightly larger sequence $\Lambda' = \{\lambda'_n\} \supset \Lambda$ that satisfies $\sup_n(\lambda'_n - \lambda'_{n-1}) \leq C$ and

$$\inf_{\lambda'_n, \lambda'_{n-1} \in \Lambda' \setminus \Lambda} (\lambda'_n - \lambda'_{n-1}) \geq C/2.$$

Since $C_\Lambda < 1$, for large enough C the sequence Λ' will still satisfy $C_{\Lambda'} < 1$.

The inner function θ should then be chosen for the sequence Λ' instead of Λ . Consider the outer function

$$h = (1 - \theta)K\nu \in K_\theta^{1,\infty}.$$

Then h is divisible by S and has zeros at $\Upsilon = \Lambda' \setminus \Lambda$. Since Υ is a separated sequence, there exists an inner function I , $\text{spec}_I = \Upsilon$ such that $(\arg I)'$ is bounded, see for instance lemma 16 in [7]. If C is large enough, $|(\arg I)'| < \varepsilon$. The function

$$g = \frac{Ih}{1 - I}$$

is divisible by S and satisfies

$$\bar{\theta}g = \bar{\theta} \frac{Ih}{1 - I} = \bar{\theta}h \frac{1}{1 - I}$$

on $\mathbb{R}$. Therefore $g \in K_\theta^+$. At the same time, g no longer has zeros on $\mathbb{R}$. Denote $f = g/IS$. Then $f \in N^+[\bar{\theta}S]$ is an outer function whose argument on $\mathbb{R}$ is equal to $(\arg \theta - x - \arg I)/2$. Now we can apply claim 1 to $u = \arg \theta - x - \arg I$ to obtain functions $v = v_1 + v_2$ satisfying the properties 1-5.

If one denotes by Γ_n the middle one-third of $(\lambda'_n, \lambda'_{n+1})$, then similarly to (6.28),

$$|S^{-1}(x)h(x)| = |(1 - \theta(x))K\nu(x)| \leq 6\|\nu\| |\lambda'_{n+1} - \lambda'_n|^{-1}.$$

The argument of the function h/S is $\arg \theta - x$. Note that claim 3 still holds with Λ' in place of Λ , because Υ is separated. Hence (6.29) still holds for $\phi = \arg \theta - x$.

After that, using the property that $|(\arg I)'| < \varepsilon$, one can "absorb" $\arg I$ into v_1 and replace v_1 with $y = v_1 + \arg I$. The rest of the estimates of the integral in (6.25) can be done in the same way as before, with $v = y + v_2$ in place of $v = v_1 + v_2$. $\square$

7. USED TECHNICALITIES

This section contains several lemmas and corollaries used in previous sections.

If Λ is a real sequence we define its Beurling-Malliavin density as

$$d_{BM}(\Lambda) = \sup\{d \mid \exists \text{ long } \{I_n\} \text{ such that } \#\Lambda \cap I_n \geq d|I_n| \forall n\}$$

if Λ is discrete and as ∞ otherwise.

An equivalent definition is given in [19]:

$$d_{BM}(\Lambda) = \sup\{a : N[\bar{S}^{2\pi a}\theta] = 0\},$$

where $\theta(z)$ denotes some/any meromorphic inner function with $\text{spec}_\theta = \Lambda$.

Note that the Beurling-Malliavin multiplier theorem implies that $N[\bar{S}^{2\pi a}\theta]$ in the above definition can be replaced with any $N^p[\bar{S}^{2\pi a}\theta]$, $0 < p \leq \infty$, the kernel in the Hardy space H^p , or with $N^+[\bar{S}^{2\pi a}\theta]$, the kernel in the Smirnov class.

Lemma 1. *Let $X \subset \mathbb{R}$ be a closed set and let Λ be a discrete sequence. Then*

$$G_{X \cup \Lambda} \leq G_X + 2\pi d_{BM}(\Lambda).$$

Proof. Let $d_{BM}(\Lambda) = d_1$, $G_X = d_2$ and $G_{X \cup \Lambda} = d_3$. Let $\varepsilon > 0$ be a small number. By theorem 1, $N[\theta S^{d_3 - \varepsilon}] \neq 0$ for some meromorphic inner θ , $\text{spec}_\theta \subset X \cup \Lambda$. Let $f \in N[\bar{\theta} S^{d_3 - \varepsilon}]$. Let I be an inner function such that $\text{spec}_I = \Lambda$.

By the above definition of the Beurling-Malliavin density, there exists a function $g \in N^\infty[\bar{S}^{2\pi d_1 + \varepsilon} I]$. Then the function $h = (1 - I)g$ belongs to $N^\infty[\bar{S}^{2\pi d_1 + \varepsilon}]$ and is equal to 0 on Λ . The function fh belongs to $N[\bar{\theta} S^{d_3 - 2\pi d_1 - 2\varepsilon}]$ and is zero on Λ (obviously, we assume that $d_3 - 2\pi d_1 - 2\varepsilon > 0$). Finally, the function $l = S^{d_3 - 2\pi d_1 - 2\varepsilon} fh$ belongs to $N[\bar{\theta}] = K_\theta$ and is still zero on Λ . By the Clark representation

$$l = \frac{1}{2\pi i} (1 - \theta) K l \sigma$$

where σ is the Clark measure for θ , $\text{supp } \sigma = \text{spec}_\theta \subset \Lambda \cup X$. Since l is divisible by $S^{d_3 - 2\pi d_1 - 2\varepsilon}$ in $\mathbb{C}_+$ and $(1 - \theta)$ is an outer function in $\mathbb{C}_+$, $K l \sigma$ is divisible by $S^{d_3 - 2\pi d_1 - 2\varepsilon}$ in $\mathbb{C}_+$. Equivalently, the measure $l \sigma$ has a spectral gap of the size $d_3 - 2\pi d_1 - 2\varepsilon$. Since l is zero on Λ , the measure $l \sigma$ is supported on X . Hence

$$G_X \geq d_3 - 2\pi d_1 - 2\varepsilon = G_{X \cup \Lambda} - 2\pi d_{BM}(\Lambda) - 2\varepsilon.$$

□

The following statement can be viewed as a version of the first Beurling-Malliavin theorem, see [19, 20].

Lemma 2. *Let Λ be a real sequence. Suppose that there exists a long sequence of intervals I_n such that*

$$\#(\Lambda \cap I_n) \leq a|I_n| \tag{7.1}$$

for all n , for some $a \geq 0$. Then $G_\Lambda \leq 2\pi a$.

Proof. Suppose that $G_\Lambda = 2\pi a + 3\varepsilon$ for some $\varepsilon > 0$. Then by theorem 1, $N[\bar{\theta}S^{2\pi a + 2\varepsilon}] \neq 0$ for some inner function θ , $\text{spec}_\theta \subset \Lambda$. But (7.1) implies that the argument of the symbol increases greatly on I_n , which leads to a contradiction. More precisely, denote

$$\gamma = \arg(\bar{\theta}S^{2\pi a + 2\varepsilon}) = (2\pi a + 2\varepsilon)x - \arg \theta.$$

For each $I_n = (a_n, a_{n+1}]$ denote

$$\delta_n = \inf_{I_n''} \gamma - \sup_{I_n'} \gamma,$$

where

$$I_n' = (a_n, a_n + \frac{\varepsilon|I_n|}{6(\pi a + \varepsilon)}) \quad \text{and} \quad I_n'' = (a_{n+1} - \frac{\varepsilon|I_n|}{6(\pi a + \varepsilon)}, a_{n+1}).$$

Then (7.1) implies that $\delta_n \geq \frac{\varepsilon}{3}|I_n|$. Hence by a theorem in [19], section 4.4, $N[\bar{\theta}S^{a+2\varepsilon}]$ has to be trivial. $\square$

Lemma 3. *Let $I = [a, b]$ be an interval on $\mathbb{R}$ and let $\Lambda = \{\lambda_1, \dots, \lambda_N\}$, $a \leq \lambda_1 < \dots < \lambda_N \leq b$ be a set of points on I . Let $C > 1$ be a constant and suppose that for some subinterval $J = [c, d] \subset I$,*

$$\#\Lambda \cap J \leq \frac{|J|}{C} - 1.$$

Then one can spread the points of Λ on J without a large decrease in the energy $E(\Lambda)$. More precisely, if

$$\Gamma = \{\gamma_1, \dots, \gamma_N\}, \quad a \leq \gamma_1 < \dots < \gamma_N \leq b$$

is another set of points on I with the properties that

- 1) $\gamma_k = \lambda_k$ for all k such that $\lambda_k \notin J$ and
- 2) $|\gamma_k - \gamma_j| \geq C$ for all $\gamma_k, \gamma_j \in J$, $\gamma_k \neq \gamma_j$

then

$$E(\Gamma) \geq E(\Lambda) - \frac{\log C}{C}|J|N.$$

where E is defined by (5.1)

Proof. Notice that $\sum_{\gamma_k, \gamma_j \in J} \log_- |\gamma_k - \gamma_j| = 0$ and that

$$\log_+ |\gamma_k - \gamma_j| \geq \log_+ |\lambda_k - \lambda_j| - \log_+ C$$

for all k, j . $\square$

Corollary 1. *Let Λ be a sequence of real points that satisfies the density (5.3) and energy (5.4) conditions for some partition I_n and $d > 0$. Let $C > 1$. Let J_k be a sequence of disjoint intervals such that for every k , $J_k \subset I_n$ for some n and*

$$\#\Lambda \cap J_k \leq \frac{|J_k|}{C} - 1$$

for all k . Let Γ be a sequence of points obtained from Λ by spreading the points on each interval J_k like in the last lemma. Then Γ satisfies the density and energy conditions with the same partition I_n and d .

Corollary 2. *Let $\Lambda = \{\lambda_n\}$ be a monotone sequence of real points such that $C_\Lambda \geq d > 0$. Then for any $\varepsilon > 0$ there exists a monotone sequence $\Gamma = \{\gamma_n\}$ such that*

- 1) $C_\Gamma \geq d$,
- 2) $d_{BM}(\Gamma \setminus \Lambda) < \varepsilon$ and
- 3) $\sup_n(\gamma_{n+1} - \gamma_n) < \infty$.

Proof. Choose $C > 0$ so that $1/C \ll d$ and $1/C \ll \varepsilon$. Let $[\lambda_{n_k}, \lambda_{n_k+1}]$ be a sequence of all "gaps" of Λ satisfying $\lambda_{n_k+1} - \lambda_{n_k} > C$.

Since $C_\Lambda \geq d$, there exists a partition I_n such that Λ satisfies (5.3) and (5.4) for I_n and d . One can choose a sequence of disjoint intervals J_k such that for every k , $J_k \subset I_n$ for some n ,

$$\cup[\lambda_{n_k}, \lambda_{n_k+1}] \subset \cup J_k \quad \text{and} \quad \frac{|J_k|}{2C} \leq \#\Lambda \cap J_k \leq \frac{|J_k|}{C} - 1 \quad \text{for all } k$$

(the choice of J_k can be made by a version of the "shading" algorithm, see for instance [14]), volume 2, pp 507–508. Let Γ be a sequence of points obtained from Λ by spreading the points on each interval J_k like in lemma 3. Then 1) is satisfied by the previous corollary and the supremum in 3) is at most $2C$. Since the distances between the points of Γ on $\cup J_k$ are at least C ,

$$d_{BM}(\Gamma \setminus \Lambda) \leq C^{-1} < \varepsilon.$$

□

Lemma 4. *Let Λ be a sequence of real points and let $\{I_n\}$ be a short partition such that Λ satisfies*

$$a|I_n| < \#\Lambda \cap I_n$$

for all n with some $a > 0$ and the energy condition (5.4) on $\{I_n\}$. Then for any short partition $\{J_n\}$, there exists a subsequence $\Gamma \subset \Lambda$ that satisfies

$$\#(\Lambda \setminus \Gamma) \cap J_n = o(|J_n|)$$

as $n \rightarrow \pm\infty$, and the energy condition (5.4) on $\{J_n\}$.

Proof. To simplify the estimates we will assume that the endpoints of I_n belong to Λ , i.e. that $I_n = (\lambda_{k_n}, \lambda_{k_n+1}]$ for each n , and that the energy condition (5.4) is satisfied on I_n with E_n defined by (6.1), see the explanation there.

(To include the endpoints in E_n one may need to compensate by deleting a point on each I_n , as explained in the beginning of the proof of theorem 2. This is where one may need to pass from Λ to a subsequence Γ . Since $|I_n| \rightarrow \infty$, Γ will satisfy $\#(\Lambda \setminus \Gamma) \cap J_n = o(|J_n|)$.)

We will also assume that $\#\Lambda \cap I_n = |I_n|$ for all n . In this case one can choose $\Gamma = \Lambda$.

Fix n and suppose that the intervals $I_l, \dots, I_{l+N}$ cover J_n . To estimate the energy expression for J_n let us first consider the case when $\cup_{l=l_0}^{l_0+N} I_l = J_n$. Denote by

u a piecewise linear continuous function on $\mathbb{R}$ that is zero outside of J_n and grows linearly by 1 between λ_n and λ_{n+1} for each $\lambda_n, \lambda_{n+1} \in J_n$. Denote

$$p(x) = \begin{cases} u(x) - x + \lambda_{k_l} & \text{on } J_n = (\lambda_{k_l}, \lambda_{k_l+N+1}] \\ 0 & \text{on } \mathbb{R} \setminus J_n \end{cases}.$$

Then $p(\lambda_{k_n}) = 0$ for all $l \leq n \leq l + N + 1$. Denote by p_n the restriction $p_n|_{I_n}$.

On each $(\lambda_i, \lambda_{i+1})$ the function u' satisfies the same estimates as $|\theta'|$ from the statement of lemma 5, parts 2 and 3. Therefore for the function p one can apply the same argument as in the first part of the proof of theorem 2, where p was defined as $\arg \theta - x - v_2$ (we will simply assume that $v_2 \equiv 0$).

First, one can show that

$$\begin{aligned} & - \iint_{J_n \times J_n} \log |t - x| p'(t) p'(x) dt dx = \\ & |J_n|^2 \log |J_n| - \sum_{\lambda_{k_l} \leq \lambda_i, \lambda_j \leq \lambda_{k_l+N+1}, \lambda_i \neq \lambda_j} \log |\lambda_i - \lambda_j| + \text{const} |J_n|^2. \end{aligned}$$

To estimate the last integral rewrite it as

$$- \iint_{J_n \times J_n} \log |t - x| p'(t) p'(x) dt dx = \sum_{I_i \subset J_n} \sum_{I_j \subset J_n} - \iint_{I_i \times I_j} \log |t - x| p'(t) p'(x) dt dx.$$

For the last integral, when $i = j$ by (6.16) and (6.17) we have

$$- \iint_{I_i \times I_i} \log |t - x| p'(t) p'(x) dt dx \lesssim |I_i|^2 \log |I_i| - E_i + \text{const} |I_i|^2.$$

If I_i does not intersect $2I_j$ we can apply (6.18) and (6.19) (where we used that $\text{dist}(I_j, I_k) \geq |I_j|$) to obtain

$$- \iint_{I_i \times I_j} \log_+ |t - x| p'(t) p'(x) dt dx \lesssim |I_i| |I_j|.$$

For the case when I_i intersects $2I_j$ but not contained in $2I_j$, or when I_i is adjacent to I_j , (note that there are at most four of such I_i for each I_j) we can estimate the integral like in (6.20) to conclude that

$$\begin{aligned} & - \iint_{I_i \times I_j} \log_+ |t - x| p'(t) p'(x) dt dx \lesssim \\ & (|I_i|^2 \log |I_i| - E_i) + (|I_j|^2 \log |I_j| - E_j) + |I_i|^2 + |I_j|^2. \end{aligned}$$

Finally, in the case when $I_i \subset 2I_j$, $j > i + 1$, notice that for any $x \in I_i$ and any $s, t \in I_j$, $t > s$, $\log_+ |s - t| < \log_+ |x - t|$. Hence (we assume that $\text{dist}(x, I_j) \geq |I_{i+1}| > 1$ to skip the estimates of $\log_-$)

$$\begin{aligned} & - \iint_{I_j \times I_j} \log_+ |s - t| p'(t) p'(s) dt ds = -2 \int_{\lambda_{k_j}}^{\lambda_{k_j+1}} \int_s^{\lambda_{k_j+1}} \log_+ |s - t| p'(t) p'(s) dt ds = \\ & 2 \int_{\lambda_{k_j}}^{\lambda_{k_j+1}} \int_s^{\lambda_{k_j+1}} \log_+ |s - t| dt ds - 2 \int_{\lambda_{k_j}}^{\lambda_{k_j+1}} \int_s^{\lambda_{k_j+1}} \log_+ |s - t| u'(t) u'(s) dt ds \geq \end{aligned}$$

$$2 \left(|I_j|^2 \log |I_j| - |I_j| \int_{I_j} \log_+ |x-t| u'(t) dt \right) + \text{const} |I_j|^2.$$

Also for any $x \in I_i$

$$\int_{I_j} \log_+ |x-t| dt - \int_{I_j} \log_+ |x-t| u'(t) dt \gtrsim -|I_j|.$$

Therefore

$$\begin{aligned} & - \iint_{I_i \times I_j} \log_+ |t-x| p'(t) p'(x) dt dx \leq \\ & \int_{I_i} |p'(x)| \left(\int_{I_j} \log_+ |x-t| dt - \int_{I_j} \log_+ |x-t| u'(t) dt + \text{const} |I_j| \right) dx \leq \\ & 2|I_i| \left(|I_j| \log |I_j| - \int_{I_j} \log_+ |x-t| u'(t) dt \right) + \text{const} |I_i| |I_j| \leq \\ & - \frac{|I_i|}{|I_j|} \iint_{I_i \times I_j} \log_+ |s-t| p'(t) p'(s) dt ds + \text{const} |I_i| |I_j| = \\ & \frac{|I_i|}{|I_j|} \|p_j\|_{\mathcal{D}} + \text{const} |I_i| |I_j| \lesssim \\ & \frac{|I_i|}{|I_j|} (|I_j|^2 \log |I_j| - E_j) + |I_i| |I_j|. \end{aligned}$$

Combining the estimates and using the shortness of $\{J_n\}$, we obtain that Λ satisfies the energy condition on $\{J_n\}$.

In the case when the intervals $I_l, \dots, I_{l+N}$ cover J_n but $\cup_l^{l+N} I_j \neq J_n$, i.e. when $I_l, I_{l+N} \cap J_n \neq \emptyset$ but at least one of I_l, I_{l+N} is not a subset of J_n , denote $I_l^* = I_l \cap J_n$ and $I_{l+N}^* = I_{l+N} \cap J_n$. Notice that by remark 3 and the fact that $\log |I_l^*| < \log |I_l|$,

$$|I_l^*|^2 \log |I_l^*| - E_l^* \leq |I_l|^2 \log |I_l| - E_l.$$

Similarly,

$$|I_{l+N}^*|^2 \log |I_{l+N}^*| - E_{l+N}^* \leq |I_{l+N}|^2 \log |I_{l+N}| - E_{l+N}.$$

Now we can use the previous case with I_l^*, I_{l+N}^* in place of I_l, I_{l+N} . □

Corollary 3. *Let Λ be a sequence of real points and let $\{J_n\}$ be a short partition such that Λ satisfies the density condition (5.3) with some $a > 0$ and the energy condition (5.4). Then for any $\varepsilon > 0$ there exists a subsequence $\Gamma \subset \Lambda$ and a short monotone partition J_n such that Γ satisfies (5.3), with $d - \varepsilon$ in place of d , and (5.4) on J_n .*

Proof. One can choose a short monotone partition $\{J_n\}$ satisfying

$$(a - \varepsilon) |J_n| \leq \#\Lambda \cap J_n$$

for all n . Such a partition can be constructed in the same way as in the second part of the proof of theorem 2, see the second paragraph before claim 3. Then Γ can be found by lemma 4. □

Lemma 5. *Let $A = \{a_n\}_{n \in \mathbb{Z}}$ be a real sequence satisfying*

$$a_n < a_{n+1}, \quad a_{n+1} - a_n < C < \infty$$

and $a_n \rightarrow \pm\infty$ as $n \rightarrow \pm\infty$. Denote $I_n = (a_n, a_{n+1})$. Then there exists an inner function θ satisfying

- 1) $\text{spec}_\theta = A$,
- 2) $|I_n|^{-1} \lesssim |\theta'| \lesssim |I_n|^{-2}$ on the middle one-third of I_n , for all n ;
- 3) $|\theta'| \lesssim [\min(|I_{n-1}|, |I_n|, |I_{n+1}|)]^{-1}$ on the rest of I_n , for all n .

Proof. Define a second sequence B as the sequence of midpoints of complimentary intervals of A in $\mathbb{R}$: $b_n = (a_n + a_{n+1})/2$.

Define the inner function θ to satisfy

$$\frac{1 - \theta}{1 + \theta} = \text{const } e^{Ku}, \quad (7.2)$$

where $u = 1_E - 1/2$,

$$E = \bigcup_{k=-\infty}^{\infty} (a_k, b_k),$$

and Ku is the *improper* integral

$$Ku(z) = \int \frac{u(t) dt}{t - z}, \quad (z \in \mathbb{C}_+).$$

The integral converges since u is a convergent sum of atoms $u|_{[a_n, a_{n+1}]}$.

(Formulas similar to (7.2) are often used in perturbation theory. In those settings, u is the Krein-Lifshits shift function and θ is the characteristic function of the perturbed operator, see for instance [25, 28])

Let μ_1, μ_{-1} be the Clark measures for θ defined by the Herglotz representation

$$\begin{aligned} \frac{1 + \theta}{1 - \theta} &= \frac{1}{\pi i} \int_{\mathbb{R}} \left[\frac{1}{t - z} - \frac{t}{1 + t^2} \right] dt \mu_1(t) + \text{const}, \\ \frac{1 - \theta}{1 + \theta} &= \frac{1}{\pi i} \int_{\mathbb{R}} \left[\frac{1}{t - z} - \frac{t}{1 + t^2} \right] dt \mu_{-1}(t) + \text{const}. \end{aligned}$$

The measures μ_1, μ_{-1} have the following form:

$$\mu_1 = \sum \alpha_n \delta_{a_n}, \quad \mu_{-1} = \sum \beta_n \delta_{b_n}$$

for some positive numbers α_n, β_n . (It is easy to see that $\mu_{\pm 1}\{\infty\} = 0$ although we don't actually need this fact.)

Put $\delta_n = a_{n+1} - a_n$. We claim that

$$\delta_n^2 \lesssim \beta_n \lesssim \delta_n. \quad (7.3)$$

Assuming that this estimate holds, we could finish as follows. Since

$$|\theta'| \asymp |1 - \theta|^2 |(\mathcal{S}\mu_1)'|, \quad |\theta'| \asymp |1 + \theta|^2 |(\mathcal{S}\mu_{-1})'|,$$

we have

$$|\theta'(x)| \asymp \min \left\{ \sum \frac{\alpha_n}{(x - a_n)^2}, \sum \frac{\beta_n}{(x - b_n)^2} \right\}, \quad (x \in \mathbb{R}).$$

Now if x belongs to the middle one-third of one of the intervals (a_m, a_{m+1}) , then $|\theta'(x)|$ can be estimated as

$$|\theta'(x)| \asymp \sum \frac{\alpha_n}{(x - a_n)^2} \asymp \sum \frac{\alpha_n}{(b_m - a_n)^2} = \beta_m^{-1}$$

and the estimate follows from (7.3). On the rest of the interval $|\theta'(x)|$ can be estimated by $\sum \frac{\beta_n}{(x - b_n)^2}$ which together with the right half of (7.3) gives the desired estimate.

It remains to prove (7.3). As follows from (7.2),

$$\beta_n = \text{const Res}_{b_n} e^{-Ku}.$$

Denote

$$g_n(z) = \exp \left\{ - \int_{a_n}^{a_{n+1}} \frac{u(t) dt}{t - z} \right\} = \frac{\sqrt{(a_n - z)(a_{n+1} - z)}}{b_n - z},$$

and

$$A_n = \exp \left\{ - \int_{\mathbb{R} \setminus (a_n, a_{n+1})} \frac{u(t) dt}{t - b_n} \right\},$$

so

$$\text{Res}_{b_n} e^{-Ku} = A_n \text{Res}_{b_n} g_n, \quad |\text{Res}_{b_n} g_n| = \frac{1}{2} \delta_n.$$

To prove the right half of (7.3) notice that $A_n \lesssim 1$. Indeed, to the right from a_{n+1} , on each (a_j, a_{j+1}) the function u is positive on the half of the interval that is closer to b_n and negative on the half that is further from it. Thus

$$- \int_{(a_{n+1}, \infty)} \frac{u(t) dt}{t - b_n} < 0.$$

Similarly

$$\int_{(-\infty, a_n)} \frac{u(t) dt}{t - b_n} < 0.$$

To prove the left half of (7.3) one needs to show that $\delta_n \lesssim A_n$. Notice that, since $\delta_n < C$,

$$- \sum_{\text{dist}(b_n, (a_j, a_{j+1})) \geq 1} \int_{(a_j, a_{j+1})} \frac{u(t) dt}{t - b_n} > \text{const} > -\infty.$$

As for the remaining part,

$$- \sum_{0 < \text{dist}(b_n, (a_j, a_{j+1})) \leq 1} \int_{(a_j, a_{j+1})} \frac{u(t) dt}{t - b_n} > - \int_{\delta_n/2}^{1+C} \frac{dx}{x} = \log \delta_n + \text{const}.$$

□

Our last lemma in this section can be easily verified. We state it without a proof.

Lemma 6. *Let $a_1 < a_2$ and $b_1 < b_2$ be points on the real line. Let α and β be non-negative functions on the intervals (a_1, a_2) and (b_1, b_2) correspondingly satisfying*

$$\int_{a_1}^{a_2} \alpha = \int_{b_1}^{b_2} \beta = 1, \quad \alpha < A \quad \text{and} \quad \beta < B$$

where $A, B > 1$. Then

$$1) \log_-(a_2 - a_1) \leq \int_{a_1}^{a_2} \int_{a_1}^{a_2} \log_-(x - y) \alpha(x) \alpha(y) dx dy \leq \log_- \frac{1}{A} + 1.$$

2) If $a_2 < b_1$ then

$$\log_-(b_2 - a_1) \leq \int_{a_1}^{a_2} \int_{b_1}^{b_2} \log_-(x - y) \alpha(x) \beta(y) dx dy \leq \log_-(b_1 - a_2).$$

3) If $a_2 = b_1$ then

$$\int_{a_1}^{a_2} \int_{b_1}^{b_2} \log_-(x - y) \alpha(x) \beta(y) dx dy \leq \min(\log_- \frac{1}{A}, \log_- \frac{1}{B}) + 1.$$

4) If $a_2 \leq b_1$ then

$$\log_+(b_1 - a_2) \leq \int_{a_1}^{a_2} \int_{b_1}^{b_2} \log_+(x - y) \alpha(x) \beta(y) dx dy \leq \log_+(b_2 - a_1).$$

5) If $A/2 \leq \alpha(x) \leq A$ on (a_1, a_2) then for any $y \in (a_1, a_2)$

$$\log_+ |a_2 - a_1| - C \leq \int_{a_1}^{a_2} \log_+(x - y) \alpha(x) dx \leq \log_+ |a_2 - a_1| + C$$

for some absolute constant C .

6) If $A/2 \leq \alpha(x) \leq A$ on (a_1, a_2) then for any $y > a_2$

$$\log_+ |y - a_1| - C \leq \int_{a_1}^{a_2} \log_+(x - y) \alpha(x) dx \leq \log_+ |y - a_1|$$

for some absolute constant C .

8. DE BRANGES' THEOREM 66 IN THE TOEPLITZ FORM

In this section we discuss a Toeplitz version of theorem 66 from [7]. For reader's convenience we include the proofs.

We say that a finite measure μ on $\mathbb{R}$ annihilates K_θ if $\int f d\mu = 0$ for a dense set of $f \in K_\theta$. Note that the integral always exists for a dense set of functions since, for instance, the disk algebra is dense in every K_θ .

We say that the Cauchy integral $K\mu$ is divisible by an inner function θ if $K\mu/\theta = K\eta$ in $\mathbb{C} \setminus \mathbb{R}$ for some finite complex measure η on $\mathbb{R}$.

Lemma 7. [1] *Let μ be a finite complex measure on $\mathbb{R}$ and let θ be an inner function in $\mathbb{C}_+$. Then the following statements are equivalent:*

- (i) μ annihilates K_θ ;
- (ii) The Cauchy integral of the conjugate measure $\bar{\mu}$, $K\bar{\mu}$, is divisible by θ .

Proof. (i) $\Rightarrow$ (ii). We will assume that the reproducing kernels of K_θ belong to the dense set annihilated by μ (otherwise one needs to use a standard limiting procedure). If $\lambda \in \mathbb{C}_+$ then

$$0 = \overline{\int \frac{1 - \bar{\theta}(\lambda)\theta(z)}{\bar{\lambda} - z} d\mu(z)} = \theta(\lambda)K\bar{\theta}\bar{\mu}(\lambda) - K\bar{\mu}(\lambda)$$

which implies the statement.

(ii) $\Rightarrow$ (i). Let J be the inner function whose Clark measure is $|\mu|$. Then the function $f = (1 - J)K\bar{\mu}$ belongs to K_J and is divisible by θ . Let $f = \theta g$. We will assume that $\theta(i) = 0$. The general case can be treated similarly.

Let

$$\theta = \theta^* \frac{z - i}{z + i}.$$

If I is an inner divisor of θ^* then $Ig/(z + i)$ also belongs to K_J and is summable on the real line. Recall that the integral over $\mathbb{R}$ of a function from $H^1(\mathbb{C}_+)$ is 0. At the same time, by the Clark representation, for each function h from K_J ,

$$\int_{\mathbb{R}} h(x) dx = \int h(x) d|\mu|(x).$$

Hence

$$0 = \int \frac{(Ig)(x)}{x + i} dx = \int \frac{(Ig)(x)}{x + i} d|\mu|(x) = \int \frac{I(x)\bar{\theta}(x)f(x)}{x + i} d|\mu|(x) = \overline{\int \frac{\theta\bar{I}}{x - i} d\mu}.$$

Since functions $\frac{\theta}{I(z - i)}$ are complete in K_θ , we obtain the statement. $\square$

Like in de Branges' proof of theorem 66, we will use the Krein-Milman theorem on the existence of extreme points in a convex set to obtain the following lemma:

Lemma 8. *Let θ be an inner function in $\mathbb{C}_+$. Let μ be a finite complex measure whose Cauchy integral $K\mu$ is divisible by θ (or, equivalently, $\bar{\mu}$ annihilates K_θ). Then there exists a finite complex measure ν such that*

- 1) $\text{supp } \nu \subset \text{supp } \mu$;
- 2) $K\nu$ is divisible by θ ($\bar{\nu}$ annihilates K_θ);
- 3) $K\nu$ has no zeros outside of $\text{supp } \nu$, except the zeros of θ in $\mathbb{C}_+$;
- 4) if θ is a meromorphic inner function then ν is concentrated on a discrete set.

Proof. First, let us symmetrize μ . Since together with any $f \in K_\theta, \theta\bar{f} \in K_\theta$, the measure $\bar{\theta}\mu$, just like $\bar{\mu}$, annihilates K_θ and $K\bar{\theta}\mu$ is divisible by θ . Consider $\eta = \mu + \theta\bar{\mu}$. WLOG $\|\eta\| \leq 1$.

Denote $\Sigma = \text{supp } \mu$. Let A_Σ^θ be the set of all finite complex measures σ such that $\|\sigma\| \leq 1$, $\text{supp } \sigma \subset \Sigma$, the Cauchy integral of σ is divisible by θ and

$$\theta\bar{\sigma} = \sigma. \tag{8.1}$$

Since $\eta \in A_\Sigma^\theta$, this set is not empty. It is also convex. By the Krein-Milman theorem it contains a non-zero extremal point ν . We claim that this is the desired measure.

First, let us show that the set of real $L^\infty(|\nu|)$ -functions h such that $Kh\nu$ is divisible by θ is one-dimensional, and therefore $h = c \in \mathbb{R}$. (This is equivalent to the statement that the closure of K_θ in $L^1(|\nu|)$ has deficiency 1, i.e. the space of its annihilators is one dimensional)

Let there be a bounded real h such that $Kh\nu$ is divisible by θ . WLOG $h \geq 0$, since one can add constants, and $\|h\nu\| = 1$. Choose $0 < \alpha < 1$ so that $|\alpha h| < 1$. Consider probability measures $\nu_1 = h\nu$ and $\nu_2 = (1 - \alpha)^{-1}(\nu - \nu_1)$. Then both of them belong to A_Σ^θ and $\nu = \alpha\nu_1 + (1 - \alpha)\nu_2$ which contradicts the extremality of ν .

Now let us show that ν is a singular measure. Let g be a continuous compactly supported real function such that $\int g d\nu = 0$. By the previous part, there exists a sequence $f_n \in K_\theta$, $f_n \rightarrow g$ in $L^1(|\nu|)$ (otherwise the defect is larger than 1). Since $\bar{\nu}$ annihilates K_θ and $(f_n(z) - f_n(w))/(z - w) \in K_\theta$ for every fixed $w \in \mathbb{C} \setminus \mathbb{R}$,

$$0 = \int \frac{f_n(z) - f_n(w)}{z - w} d\bar{\nu}(z) = K f_n \bar{\nu}(w) - f_n(w) K \bar{\nu}(w)$$

and therefore

$$f_n = \frac{K f_n \bar{\nu}}{K \bar{\nu}}.$$

Taking the limit,

$$f = \lim f_n = \frac{K g \bar{\nu}}{K \bar{\nu}}.$$

Since all of f_n have pseudocontinuations, one can show that the limit function f must have one as well. Since the numerator is analytic outside the compact support of h , the measure in the denominator must be singular (Cauchy integrals of non-singular measures have jumps at the real line on the support of the a.c. part).

Moreover, f must be analytically continuable through the real line outside of clos spec_θ , like all of f_n . In particular, if θ is meromorphic, the zero set of f has to be discrete. Since ν is singular, $K\nu$ tends to ∞ at ν -a.e. point and $f = 0$ at ν -a.e. point outside of the support of g . Choosing two different g with disjoint supports we prove that if θ is meromorphic, then ν is concentrated on a discrete set.

Now let us show that $K\nu$ does not have any zeros in $\mathbb{C} \setminus \text{supp } \nu$ other than the zeros of θ . Let J be the inner function corresponding to $|\nu|$ ($|\nu|$ is the Clark measure for J). Denote $G = (1 - J)K\nu \in K_J$. Since $K\nu$ is divisible by θ and $K|\nu|$ is outer, G is divisible by θ . Let us first show that G/θ does not have an inner component in the upper half-plane. Suppose that $G = \theta UH$ for some inner U . Then $\theta(1 + U)^2 H$ also belongs to K_J . Denote

$$\gamma = \theta(1 + U)^2 H|\nu|.$$

Then $\gamma = h\nu$ for a bounded non-constant function $h = (1 + U)^2/U$. The Cauchy integral of γ is divisible by θ because $(1 - J)K\gamma = \theta(1 + U)^2 H$. We obtain a contradiction with the property that the space of annihilators is one dimensional.

Thus $G/\theta \in K_J$ is outer in C_+ . By (8.1),

$$G\bar{\theta}|\nu| = \bar{\theta}\nu = \bar{\nu}$$

and the Clark representation formula implies

$$\bar{J}G = \overline{(1 - J)K\bar{\nu}} = \overline{(1 - J)KG\bar{\theta}|\nu|} = \overline{G/\theta},$$

so the pseudocontinuation of G does not have zeros in $\mathbb{C}_-$.

If G has a zero at $x = a \in \mathbb{R}$ outside of spec_J then

$$\frac{G}{x-a} \in K_J$$

and the measure

$$\gamma = \frac{G}{x-a} |\nu|$$

leads to a similar contradiction with the property that the space of annihilators is one-dimensional, since $(x-a)^{-1}$ is bounded on the support of ν . Since

$$G = (1-J)K\nu \in K_J,$$

$K\nu$ does not have any extra zeros. $\square$

Our last statement is a Toeplitz version of theorem 66. Recall that a function $f \in N[\phi]$ is called purely outer if f is outer in the upper half-plane and $\phi f = \bar{g}$ is outer in the lower half plane.

Corollary 4. *Let I, θ be inner functions in $\mathbb{C}_+$. Suppose that the kernel $N[\bar{I}\theta]$ is non-trivial.*

Then there exists an inner function J in $\mathbb{C}_+$ such that $\text{spec}_J \subset \text{spec}_I$ and the kernel $N[\bar{J}\theta]$ contains a purely outer function f that does not have any zeros on $\mathbb{R} \setminus \text{spec}_J$. If σ_1 is the Clark measure of J then f is also non-zero σ_1 -a.e. on spec_J .

If θ is a meromorphic function, then J can be chosen as a meromorphic function.

Proof. In the statement of lemma 8, consider I to be the inner function whose Clark measure is $|\mu|$ and let J be the inner function whose Clark measure is $|\nu|$. $\square$

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